

Null subjects, labeling, and Linearization

Microvariation in subject and agreement drop

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Abstract

Theoretical approaches to null subjects standardly focus on the narrow syntactic module. I offer a postsyntactic alternative in terms of PF deletion applying to salvage an unlinearizable derivation. Support for this alternative comes from microvariation in three closely-related West Slavic pro-drop languages spoken primarily in Poland (Kashubian, Polish, and Silesian), in which the distribution of null subjects is sensitive to the postsyntactic realization of [PERS/NUM] agreement. I argue that this microvariation indicates that in these languages at least, the licensing of null subjects takes place at PF, which in turn suggests that PF has a bigger role to play in a general theory of null arguments than has so far been assumed.

Keywords

pro-drop, agreement, rescue by PF deletion, null arguments, Linearization, Slavic languages

1 Introduction

In the original conceptions of the Null Subject Parameter (Chomsky 1981, 1982; Rizzi 1982, 1986), the licensing of a null subject applies in the syntactic module of the grammar. While over the years the formulation of the licensing conditions has been developed and transformed in a number of illuminating ways, this intuition has remained largely unchanged. Recent studies of consistent null subject languages (which the Null Subject Parameter was originally intended for) continue to focus on the narrow syntactic side of the derivation (see, e.g., Barbosa 1995, 2011, 2019; Holmberg 2005, 2010; Frascarelli 2007, 2018; Cole 2010; Roberts 2010, 2019; Sigurðsson 2011; Camacho 2013; Koenenman and Zeijlstra under revision). In fact, for some accounts (Frascarelli 2007, 2018; Holmberg 2010; Sigurðsson 2011), achieving identification is crucially dependent on syntactic processes devoid of any morphosyntactic reflexes (broadly speaking, Agree operations

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linking *pro* to left peripheral, discourse-related heads).¹

In contrast to the predominant narrow syntax-centric view, in this article I argue that the PF representation of agreement ϕ -features also has a crucial role for licensing the null subject in at least some consistent null subject languages, presenting novel generalizations on microvariation in Kashubian, Polish, and Silesian (henceforth K, P, S), all closely-related West Slavic languages spoken primarily in Poland.² To preview the discussion, in section 2 I first show how one might analyse pro-drop postsyntactically by adopting Saab’s (2020; 2024) approach relying on the operation of head ellipsis at PF. Then, in section 2.1, I delve more deeply into the nature of the relevant derivations and representations. In particular, by couching the analysis within the labeling model of Chomsky (2013, 2015), it is possible to derive pro-drop from the requirement of the externalization system that hierarchical structures be linearized. I propose that PF deletion applies to pronominal subjects in consistent null subject languages to salvage a derivation whose terminals cannot be associated with unambiguous ordering statements. Section 3 then presents an overview of the K, P, and S data. I show that although K and S have rich agreement morphology and allow pro-drop in general, for certain person/number combinations, the [PERS/NUM] agreement auxiliary can or must be left unexpressed. Crucially, when this happens, pro-drop is blocked. The microvariation observed supports an analysis on which the licensing of *pro*, in these languages at least, is partly determined by the realization of the agreement features at PF. I demonstrate how exactly this is achieved for the crucial K/P/S patterns by presenting detailed derivations of each in section 3.4. Section 4 then touches upon four further, more tentative theoretical points, namely the correlation between pro-drop and T bimorphemicity, a possible extension of the approach to partial pro-drop and argument ellipsis in radical pro-drop languages, as well as the conditions on rescue by PF deletion. The results warrant the hypotheses that null subjects, and null arguments more broadly, are a consequence of the SM interface legibility conditions and that pro-drop is not a discrete phenomenon, but rather can be unified with other elliptical constructions as an instance of a PF deletion rescue mechanism, which operates to salvage derivations that would otherwise be unexternalizable. Section 5 concludes the article.

2 Licensing *pro* at PF

The primary division which I focus on in this article is between non-null subject languages like English (e.g., **(She) runs*) and consistent null subject languages like Polish (e.g., *(Ona) biega* ‘she runs’). As my starting point I take Saab’s (2020; 2024) PF approach to pro-drop, which involves head ellipsis. This is an operation deleting morphosyntactic words (more specifically, their Q-

¹Throughout the text, I use the terms ‘*pro*’ and ‘pro-drop’ for null subjects and the mechanism(s) deriving them for the sake of brevity, as is customary in the literature, with no *pro*-based theoretical commitment intended.

²K and S are minority languages; K is used in Northern Poland (Pomeranian Voivodeship) and S in Southern Poland (Silesian and Opole Voivodeships) and Northern Czech Republic (Moravian-Silesian Region). In the Polish 2021 Census, 87 600 and 457 900 speakers declared K and S respectively as the main language used at home.

variables, see Saab 2020, 2022, 2024 for details) under adjacency or immediate locality with identical antecedents at PF. In particular, following the application of Agree and the movement of the subject D(P) pronoun to Spec,T in the narrow syntax in consistent null subject languages a dissociated agreement morpheme of category D is added, expanding T. Following Linearization, head ellipsis deletes the subject under adjacency with the agreement morpheme *qua* D in T (1).³

$$(1) \quad [\text{SUBJECT-D}_{[\phi]}] \prec [\text{T AGREEMENT-D}_{[\phi]} [\text{T T}_{[\text{TENSE}, \alpha]} \text{V}]] \quad [\text{output of head ellipsis (PF)}]$$

While this general approach can deliver the required results, I would now like to probe deeper into the nature of the deletion process yielding the null subject. In particular, rather than stipulating that the trigger for subject deletion at PF is just identity with an antecedent (unlike with phrasal ellipsis in the narrow syntax, which requires both identity and licensing; see, a.o., Merchant 2001; Aelbrecht 2010), I propose that the deletion is forced by the externalization requirement and belongs to the realm of rescue by deletion (see, a.o., Ross 1969; Lasnik 2001; Merchant 2001; Fox and Lasnik 2003; Bošković 2011, 2013 for influential discussions of this concept in the context of island and *that*-trace filter violations; see Mendes and Kandybowicz 2023 for a recent discussion in the context of Linearization contradictions).

2.1 *Pro-drop as rescue by deletion*

Within Chomsky’s (2013; 2015) labeling model, the label for a head/phrase structure is always provided by the head (i.e., label [X YP] = X). In the phrase/phrase configuration one of three outcomes is possible: (i) if one of the phrases is a lower copy of Internal Merge, the label is provided by the other phrase (i.e., label [~~XP~~ YP] = Y); (ii) if both are highest copies and they share a feature, the label is provided by the shared feature (i.e., label [XP_F YP_F] = ⟨F, F⟩); (iii) if the conditions in (i) or (ii) are not met, a labeling failure obtains and the derivation is unusable at the interfaces (i.e., label [XP YP] = *undefined*). Chomsky proposes that scenarios (i)–(iii) derive the effect of the EPP in languages like English. Namely, since the base position of the subject is an instance of configuration (iii), the subject needs to move to Spec,T, resulting in an instance of (ii) at the TP level and an instance of (i) at the VoiceP (a.k.a. v*P) level. In addition to the need to label VoiceP, the movement of the subject is a prerequisite for the labeling of TP in languages with ‘weak’ agreement like English, where T is a ‘weak’ head and cannot label on its own (Chomsky 2015:9). On the other hand, for pro-drop languages like Italian, whose agreement is ‘rich’, Chomsky suggests that T can label both TP and {SPEC, TP}, concluding that they lack the

³While the assumption that agreement in consistent null subject languages is of category D resonates through the literature on this language type (for various other implementations, see, esp. Barbosa 1995, 2011, 2019; Alexiadou and Anagnostopoulou 1998; Holmberg 2005, 2010; Roberts 2010, 2019), I pursue a different approach below in that I take neither agreement nor the subject to be of category D. This aligns with the approaches to nominal structures in Polish and (some) other Slavic languages according to which noun phrases in these languages lack the D head (a.o., Chierchia 1998; Willim 2000; Bošković 2012).

EPP.

2.1.1 Analysis of pro-drop in terms of labeling and Linearization

Chomsky (2013, 2015) proposes that the Labeling Algorithm applies via Minimal Search at Transfer, so that the labeled structure can be used at the interfaces. I depart from this assumption about the timing of the Labeling Algorithm and suggest that it applies when the information about the labels is actually needed. Among other advantages outlined below, this avoids the need to store the information obtained at Transfer.⁴ In particular, focusing on the SM interface and adopting Takita’s (2020) view that labeling is needed for Linearization, I suggest that the Labeling Algorithm directly precedes Linearization, proceeding cyclically (Fox and Pesetsky 2005). This makes the prediction that PF operations such as Chain Reduction can affect labeling, offering a natural explanation for why copies are ignored for this purpose (see Collins and Stabler 2016 for an alternative).

To be specific, I assume that following the narrow syntactic derivation, the structure undergoes Transfer/Spell-Out and the derivation continues at the PF interface, where postsyntactic operations such as Obliteration (radical impoverishment, i.e., deletion of the entire terminal node at PF, see, e.g., Arregi and Nevins 2012) apply, altering the representation before labeling, Linearization and Vocabulary Insertion (2) (cf., e.g., Embick and Noyer 2007; Arregi and Nevins 2012; Embick 2015).

- (2) Model assumed (PF side)
narrow syntax (Merge, Agree) → cyclic Transfer → PF/SM interface (Obliteration, Chain Reduction etc. → labeling → Linearization → Vocabulary Insertion)

Building on Fox and Pesetsky (2005), I take Linearization to proceed in two steps, the output of the first one being a set of precedence relations $LP(K)$, K a cyclic domain, and the output of the second one being a set of ordering statements for terminal nodes.⁵ I refer to the operation constructing $LP(K)$ as ‘Precede’ and, following Fox and Pesetsky (2005), I refer to the operation constructing $L(K)$ based on $LP(K)$ as ‘Linearize’ (3).

- (3) Linearization consists of two operations
1. Precede constructs the set of precedence relations $LP(K)$.
 2. Linearize constructs the set of ordering statements for terminals $L(K)$.

⁴Even though I feel that this modification has its strong merits, it is not crucial for the analysis below. An assumption that would be needed under the original conception of the timing of the Labeling Algorithm at Transfer is that Obliteration (and Chain Reduction/copy deletion), whose application in the present context is illustrated in example (34) below, removes not only the terminal node, but also the label which the terminal has provided for its set (or—in the case of copies—the entire set is removed).

⁵This is part of Spellout for Fox and Pesetsky (2005:41).

The set of precedence relations $LP(K)$ is originally defined over labeled structures, with crucial reference to the notions of the mother, specifier, and complement. I suggest to integrate this intuition into the current labeling model in the following way. First, domain K is labeled along the lines of Chomsky’s (2013; 2015) algorithm.⁶ Accordingly, instead of referring to phrase-level projections, $LP(K)$ encodes the relations in terms of heads and set labels, the latter in a head-categorial or feature format (in addition to encoding roots, irrelevant for the current discussion). Adopting the view that Vocabulary Insertion is the replacement of a Q -variable in a (terminal) head by a morphological exponent (Halle 1990; Embick 2010, 2015), I take terminal nodes to be distinguished from labels (i.e., non-terminals) by the presence of Q (and perhaps other features, on the assumption that only some features in a terminal provide a label for the set containing them; see section 4.1 below).⁷ Operating over the set $\{X_{[Q]}, \{Y_{[Q]}, \{\dots\}\}$, represented with brackets for readability in (4), Precede identifies $X_{[Q]}$, and it identifies Y as the label of $\{Y_{[Q]}, \{\dots\}\}$. Based on this, for head-initial languages like English Precede adds the statement $X_{[Q]} \prec Y$ to $LP(K)$. That is, the head precedes the set. This captures the head/complement relation.

$$(4) \quad [X \ X_{[Q]} \ [Y \ Y_{[Q]} \ \dots]] \quad \text{Precede}(X_{[Q]}, Y) = X_{[Q]} \prec Y \quad [X_{[Q]} \text{ a head, } Y \text{ a set label}]$$

Assuming that lower copies of Internal Merge undergo Chain Reduction/copy deletion and are therefore absent from the domain undergoing Linearization (see, e.g., Chomsky 1995; Nunes 2004; Fox and Pesetsky 2005; Bošković and Nunes 2007; Arregi and Nevins 2012), linearizing $\{X_{[Q]}, \{\dots\}, \{Y_{[Q]}, \{\dots\}\}$ under the movement scenario ($[XP \ YP]$ in (i) above) boils down to linearizing $\{Y_{[Q]}, \{\dots\}\}$, in accordance with the head/set pattern in (4).

For the time being, for the feature sharing scenario ($[XP_F \ YP_F]$ in (ii) above), I assume that Precede uses the relation created by Agree to order the two objects (5).⁸ In particular, for specifier-initial languages like English Precede adds a statement ordering the set containing the goal before the set containing the probe.⁹

$$(5) \quad [_{\langle F, F \rangle} [X \ X_{[F], [Q]} \ \dots] [Y \ Y_{[F], [Q]} \ \dots]] \\ \text{Precede}(X, Y) = X_{goal} \prec Y_{probe} \quad [X_{[F], [Q]} \text{ has been the goal of Agree}(Y_{[F], [Q]}, X_{[F], [Q]})]$$

⁶ Alternatively, labeling and Linearization could be interspersed, proceeding step by step from the top of the representation, a point which I abstract away from here.

⁷ Under this approach, the representation of (some) morphemes includes a Q -variable before Vocabulary Insertion. At PF Vocabulary Insertion replaces Q with a phonemic matrix (e.g., the English regular past tense morpheme *-ed* can be represented as $T_{[PAST, Q]}$ prior to Vocabulary Insertion, whose application yields $T_{[PAST, -ed]}$).

⁸Note that the label $\langle F, F \rangle$ becomes relevant only for Linearization with the higher part of the structure. I do not discuss adjuncts here.

⁹While this solution is certainly not ideal, as it requires some form of keeping track of the relations created by Agree in the narrow syntax at PF, this is needed independently on approaches to agreement such as the Distributed Agree system based on Agree-Link in the narrow syntax and Agree-Copy at PF (Arregi and Nevins 2012; Mendes et al. 2024 and related). Importantly, I do not envision a potential future improvement affecting the core of the proposal aiming at deriving pro-drop from Linearization ambiguity.

I take not only labeling, but also Linearization to be undefined when two phrases are highest copies that do not share a feature.

Regarding the second part of Linearization, that is Linearize, for concreteness I suggest that based on the information in $LP(K)$ it first creates sets of ordering statements ($L(K)$, $O(K)$, etc.) for the terminals which are not themselves included in $LP(K)$. For example, assuming the precedence relation between a head and a set $X_{[Q]} \prec Y$ in $LP(K)$, where Y is a label for the singleton set $\{Y_{[Q]}\}$ and $Y_{[Q]}$ does not feature in $LP(K)$, Linearize adds the statement $X_{[Q]} \frown Y_{[Q]}$ to $L(K)$.¹⁰ The resultant ordering statements are then ordered with respect to the terminals included in $LP(K)$ and with respect to one another with reference to the remaining information in $LP(K)$.¹¹ I furthermore assume that ordering between two heads can be determined lexically and that lexical specifications of particular heads can override the systemwide rules (cf. the notion of parametric hierarchies in Biberauer and Roberts 2012; Roberts 2012, 2019 and related).¹² My assumptions about Linearization are summarized in (6).

(6) Linearization consists of two operations

1. Precede constructs the set of precedence relations $LP(K)$ based on mutual c-command and language-specific settings.

When a head $X_{[Q]}$ and a set Y are in a mutual c-command relation, a or b apply.

- a. The relation $X_{[Q]} \prec Y$ is added to $LP(K)$. [head-initial]
 - b. The relation $Y \prec X_{[Q]}$ is added to $LP(K)$. [head-final]
- (Unless the default setting is overridden by lexical specification on $X_{[Q]}$.)

When a set X and a set Y are in a mutual c-command relation and $X_{[Q]}$ has been the goal of Agree triggered by $Y_{[Q]}$, a or b apply.

- a. The relation $X \prec Y$ is added to $LP(K)$. [specifier-initial]
- b. The relation $Y \prec X$ is added to $LP(K)$. [specifier-final]

2. Linearize constructs the set of ordering statements for terminals by introducing ordering between the terminal nodes included in $LP(K)$ and the nodes dominated by labels included in $LP(K)$.

- a. For each terminal $Y_{[Q]}$ absent in $LP(K)$ and contained in set Y , Y included in $LP(K)$, i or ii applies.
 - i. If $LP(K)$ contains $X_{[Q]} \prec Y$, $L(K)$ containing $X_{[Q]} \frown Y_{[Q]}$ is created.
 - ii. If $LP(K)$ contains $Y \prec X_{[Q]}$, $L(K)$ containing $Y_{[Q]} \frown X_{[Q]}$ is created.

¹⁰ ‘ $\frown$ ’ is a concatenation symbol indicating immediate precedence.

¹¹For the analysis of pro-drop suggested here it is sufficient to assume, following Fox and Pesetsky (2005), that a precedence relation of $X \prec Y$ simply implies that the final terminal of set X precedes the first terminal of set Y (however achieved). I adopt the specific implementation of Linearize suggested here for the purpose of explicitness.

¹² I tentatively assume that lexically-specified requirements imposing the inclusion of particular statements in $LP(K)$ may substitute (LF-vacuous) head lowering (Affix Hopping) and perhaps raising (e.g., V-to-Voice movement), though the standard accounts are also compatible with the approach to pro-drop suggested here.

- b. The resultant sets $L(K)$, $O(K)$, etc. are expanded and concatenated as dictated by the information encoded in $LP(K)$. For each terminal $X_{[Q]}$ in $L(K)$, i, ii, or iii applies successively until all terminals are included in a single set $L(K)$.
 - i. If $LP(K)$ contains $X \prec Z_{[Q]}$ or $X_{[Q]} \prec Z_{[Q]}$, $Z_{[Q]}$ is concatenated at the right edge of the string in $L(K)$.
 - ii. If $LP(K)$ contains $Z_{[Q]} \prec X$ or $Z_{[Q]} \prec X_{[Q]}$, $Z_{[Q]}$ is concatenated at the left edge of the string in $L(K)$.
 - iii. If $LP(K)$ contains $X \prec Z$, $X_{[Q]}$ a terminal in $L(K)$ and $Z_{[Q]}$ a terminal in $O(K)$, the string in $O(K)$ is concatenated at the right edge of the string in $L(K)$.

To illustrate the working of this system, I next present a simplified derivation of the sentence *She is smiling*. First, I make the standard assumption that the $[PERS(ON)]$ and $[NUM(BER)]$ agreement features reflected on the verb originate in T . I assume a pronominal representation of the subject following Ghomeshi and Massam (2020), with $[PERSON]$ and $[NUMBER]$ bundled in a single head. I take $[GENDER]$ to be contained separately in the nominalizing head n (Lowenstamm 2008; Willim 2012), the lowest functional head in the projection of the pronoun.¹³ Following Agree, the subject moves to $Spec,T$, as required on Chomsky’s (2013; 2015) assumptions. This yields the (simplified) representation in (7).¹⁴

$$(7) \quad [C_{[Q]} [[Pers/Num_{[PERS/NUM], [Q]} [n_{[GEN], [Q]}]] [T_{[TENSE], [PERS/NUM], [Q]} [Asp_{[Q]} [[Pers/Num_{[PERS/NUM], [Q]} [n_{[GEN], [Q]}]] [smile \dots]]]]]]$$

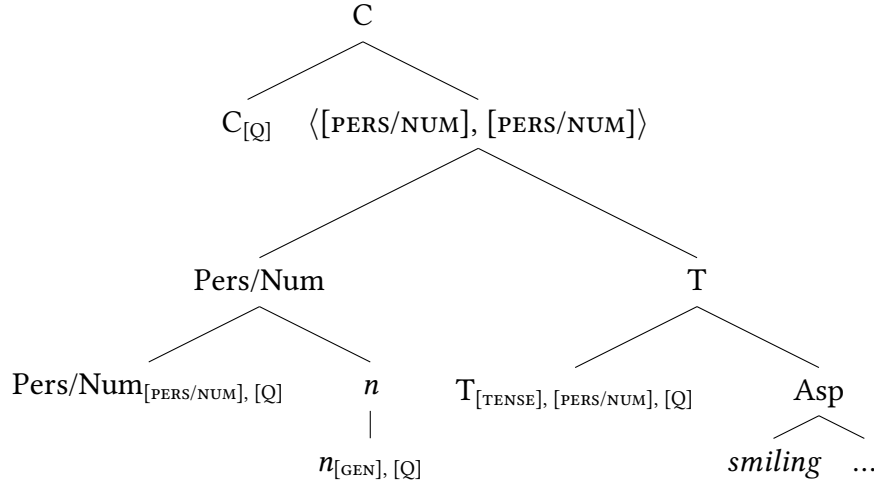
Following Chain Reduction removing the lower copy of the subject and V at PF, the Labeling Algorithm applies, yielding (8).

¹³ I assume that in constructing the pronoun Merge applies to n first, yielding the set $\{n\}$.

If the derivation can indeed start by applying Merge to the (to be the) lowest head in the structure ($Merge(X) = \{X\}$, or $Merge(X, X) = \{X\}$), contra Collins and Stabler (2016), this can be extended to roots, removing the stipulation that roots do not project because they are ‘weak’ heads (Chomsky 2013, 2015). That is, if the set constructing a noun is $\{n\{\checkmark\}\}$ rather than $\{n\checkmark\}$, labeling the noun as n follows automatically from the algorithm described in the main text, $\checkmark$ being more deeply embedded than n . See De Belder and van Craenenbroeck (2015) for some relevant discussion, as well as Moro and Roberts (2023), who achieve a similar outcome with movement.

¹⁴For clarity of exposition, I omit the details of verbal projections and represent their terminals with words rather than symbols, where irrelevant to the analysis (e.g., *smiling* in (8) is $\checkmark_{[Q]}-v_{[Q]}-Voice_{[Q]}-Asp_{[Q]}$), including the projection introducing the auxiliary in (7), which I take to originate below T (see Harwood 2014 for a discussion of different analytical options).

(8)



The labeled representation is input to the first step of Linearization, that is the operation Precede. Precede first applies to $C_{[Q]}$ and the set including the subject and the rest of the clause, which is labeled as $\langle [PERS/NUM], [PERS/NUM] \rangle$ due to the feature sharing between the Pers/Num subject and T. Since $C_{[Q]}$ is a head and $\langle [PERS/NUM], [PERS/NUM] \rangle$ is a set, the statement ordering $C_{[Q]}$ before the set is added to $LP(K)$ (9a). Precede next applies to the $\langle [PERS/NUM], [PERS/NUM] \rangle$ set, ordering the subject, as the goal of Agree, before T, the set labeled by the probe (9b). The next part of the structure along the clausal spine is linearized in accordance with the head/set rule, as in (9c). The subject pronoun is linearized as $Pers/Num_{[Q]} \prec n$ either as part of the Linearization of the CP, or separately, upon its construction, if it constitutes its own cyclic domain (9d).

(9) Precede yielding the set of precedence relations $LP(K)$ for (8)

- a. $Precede(C_{[Q]}, \langle [PERS/NUM], [PERS/NUM] \rangle) = C_{[Q]} \prec \langle [PERS/NUM], [PERS/NUM] \rangle$
 $LP(K) = \{C_{[Q]} \prec \langle [PERS/NUM], [PERS/NUM] \rangle\}$ [by head/set]
- b. $Precede(Pers/Num, T) = Pers/Num \prec T$ [by goal/probe]
 $LP(K) = \{C_{[Q]} \prec \langle [PERS/NUM], [PERS/NUM] \rangle, Pers/Num \prec T\}$
- c. $Precede(T_{[TENSE], [PERS/NUM], [Q]}, Asp) = T_{[TENSE], [PERS/NUM], [Q]} \prec Asp$ [by head/set]
 $LP(K) = \{C_{[Q]} \prec \langle [PERS/NUM], [PERS/NUM] \rangle, Pers/Num \prec T, T_{[TENSE], [PERS/NUM], [Q]} \prec Asp\}$
- d. $Precede(Pers/Num_{[Q]}, n) = Pers/Num_{[Q]} \prec n$ [by head/set]
 $LP(K) = \{C_{[Q]} \prec \langle [PERS/NUM], [PERS/NUM] \rangle, Pers/Num \prec T, T_{[TENSE], [PERS/NUM], [Q]} \prec Asp, Pers/Num_{[Q]} \prec n\}$

The final output of Precede applications shown in (9d) does not contain the $n_{[GEN], [Q]}$ and *smiling* terminals. Hence, Linearize first creates set $O(K)$ containing $Pers/Num_{[Q]} \frown n_{[Q]}$ with reference to $Pers/Num_{[Q]} \prec n$ in $LP(K)$ and set $L(K)$ containing *smiling*, by assumption with reference to suffixal lexical specifications of $v_{[Q]}$, $Voice_{[Q]}$, and $Asp_{[Q]}$ (10a). This is supplemented by adding $T_{[TENSE], [PERS/NUM], [Q]}$ to $L(K)$ (10b), at which step the sets labeled as Pers/Num and T have

both been linearized and can be concatenated based on $\text{Pers/Num} \prec T$ in $\text{LP}(K)$ (10c). Finally, $C_{[Q]}$ is added, completing Linearization of the clause (10d).¹⁵

(10) Ordering statements for terminal nodes in (8)

- a. $O(K) = \{\text{Pers/Num}_{[Q]} \frown n_{[Q]}\}$
 $L(K) = \{\sqrt{[Q]} \frown v_{[Q]}, v_{[Q]} \frown \text{Voice}_{[Q]}, \text{Voice}_{[Q]} \frown \text{Asp}_{[Q]}\} = \{\textit{smiling}\}$
- b. $O(K) = \{\text{Pers/Num}_{[Q]} \frown n_{[Q]}\}$
 $L(K) = \{T_{[\text{TENSE}], [\text{PERS/NUM}], [Q]} \frown \textit{smiling}\}$
- c. $L(K) = \{\text{Pers/Num}_{[Q]} \frown n_{[Q]} \frown T_{[\text{TENSE}], [\text{PERS/NUM}], [Q]} \frown \textit{smiling}\}$
- d. $C_{[Q]} \frown \text{Pers/Num}_{[Q]} \frown n_{[Q]} \frown T_{[\text{TENSE}], [\text{PERS/NUM}], [Q]} \frown \textit{smiling}$
 $\emptyset \quad \emptyset \quad \textit{she} \quad \textit{is} \quad \textit{smiling}$

With this background in mind, I now proceed to showing how pro-drop can be viewed within this system as an instance of rescue by deletion.

A key point of the current proposal is a formalization of the intuition that pro-drop (consistent null subject) languages are distinguished from non-pro-drop languages by ‘rich’ agreement. In particular, I suggest that the essence of the division between ‘poor’ and ‘rich’ agreement, or—in other words—‘weak’ and ‘strong’ T , lies in the contribution of the agreement ϕ -features to labeling, and—what follows—linearizing the structure. While in non-pro-drop languages the agreement ϕ -features are a subset on $T_{[\text{TENSE}], [\text{PERS/NUM}], [Q]}$, with the identity of the head and the set containing it being determined by the other feature included in it, that is $[\text{TENSE}]$, I propose that in pro-drop languages the opposite is the case, with the equivalent head effectively being a $\text{Pers/Num}_{[Q]}$ head, including the $[\text{TENSE}]$ feature.¹⁶ Viewing this from a different angle, I propose that the Labeling Algorithm targets only the dominant features in a head, that is $[\text{TENSE}]$ in non-pro-drop languages and $[\text{PERS/NUM}]$ in pro-drop languages. Put differently, there are two ways in which the features $[\text{TENSE}]$ and $[\text{PERS/NUM}]$ can be combined (in the lexicon or during the derivation on the assumption that features are combined into heads in the narrow syntax), with either one or the other contributing the label (11).¹⁷

¹⁵While nothing hinges on this, I assume here that *she* is an exponent of n in the context of singular number, with Pers/Num having a null exponent ($\emptyset$). Alternatively, one may assume that Pers/Num undergoes fusion with n .

¹⁶Note that this clausal $\text{Pers/Num}_{[Q]}$ head belongs to the traditional TP domain and is not intended to be identified with C/edge linkers in approaches such as Sigurðsson’s (2011).

¹⁷While I use the $T_{[\text{TENSE}], [\text{PERS/NUM}], [Q]}$ vs. $\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [Q]}$ notation here for clarity of exposition, this can be formalized further by taking the head including $[\text{TENSE}]$ and $[\text{PERS/NUM}]$ to be $\{[\text{TENSE}], [\text{PERS/NUM}]\}$ in non-pro-drop languages and $\{[\text{TENSE}], [\text{PERS/NUM}]\}$ in pro-drop languages (see section 4.1 below). The grammatical category of $\text{Tense}/[\text{TENSE}]$ could in fact also be absent from a hypothetical consistent pro-drop language, so long as the language has ‘rich’ subject agreement and subject movement to the projection labeled by agreement.

Parallel assumptions could likely be applied to deriving agreement-conditioned object drop in languages with ‘rich’ object agreement, though developing this idea is beyond the current scope.

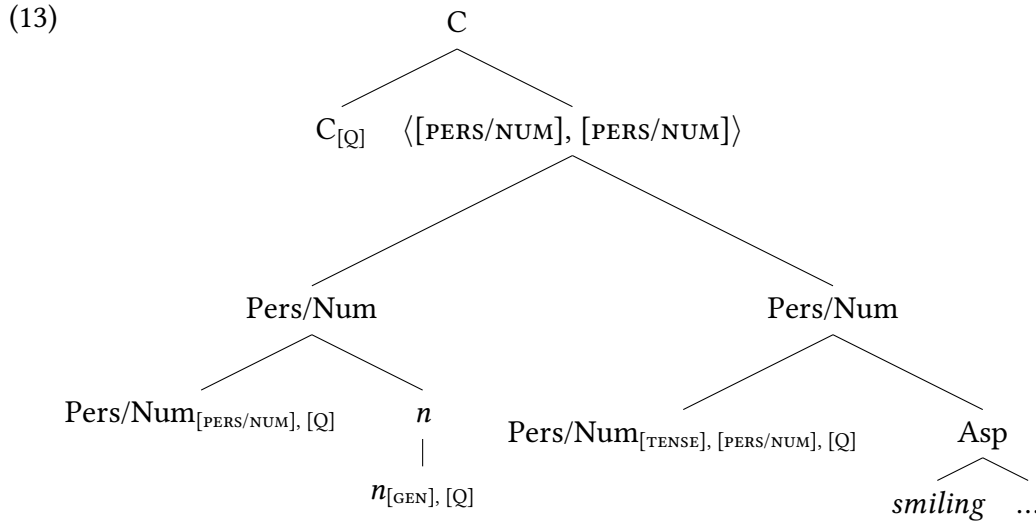
- (11) [TENSE] and [PERS/NUM] combinations into a head
- a. ‘Poor’ agreement/‘weak’ $T \rightarrow \text{non-pro-drop}$
 $T_{\{[TENSE], [PERS/NUM], [Q]\}} \rightarrow \text{Labeling Algorithm ‘sees’ } T$
 - b. ‘Rich’ agreement/‘strong’ $T \rightarrow \text{consistent pro-drop}$
 $\text{Pers/Num}_{\{[TENSE], [PERS/NUM], [Q]\}} \rightarrow \text{Labeling Algorithm ‘sees’ Pers/Num}$

These assumptions, combined with the Linearization system suggested above, offer the means to derive pro-drop from a Linearization failure (and motivate Saab’s (head) ellipsis at PF).

Consider the pro-drop version of (7) in (12), understood to belong to a hypothetical consistent null subject language, where the head containing [TENSE] and [PERS/NUM] is $\text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]}$.

- (12) a. Is smiling. ‘She is smiling.’
- b. $[C_{[Q]} [[\text{Pers/Num}_{[PERS/NUM], [Q]} [n_{[GEN], [Q]}]] [T_{[TENSE], [PERS/NUM], [Q]} [\text{Asp}_{[Q]} [[\text{Pers/Num}_{[PERS/NUM], [Q]} [n_{[GEN], [Q]}]] [\text{smile ... }]]]]]]$

As before, Chain Reduction removes the lower copy of the subject at PF and the Labeling Algorithm applies to the resultant structure (13).



The problem arises at Linearization due to the label identity of the subject and its sister (i.e., the traditional T’). Namely, Linearize cannot establish an unambiguous ordering between the subject terminals and the terminals dominated by its sister because the relevant precedence relation in LP(K) does not distinguish between the two sets at this stage. This is outlined in (14); a detailed derivation is presented immediately below.

(14) Basic intuition for a pro-drop derivation

- a. Precede adds $\text{Pers/Num} \prec \text{Pers/Num}$ to the set of precedence relations $\text{LP}(K)$ by the goal/probe relation.
- b. Linearize fails to integrate the terminals in the two sets (i.e., the subject and its sister).
- c. Deletion removes the subject set.
- d. Linearize succeeds at ordering the remaining terminals.

In a structure like (13), Precede first identifies $C_{[Q]}$ and the set including the subject and the rest of the clause, which is labeled as $\langle [\text{PERS/NUM}], [\text{PERS/NUM}] \rangle$ due to feature sharing, ordering $C_{[Q]}$ before $\langle [\text{PERS/NUM}], [\text{PERS/NUM}] \rangle$ (15a). Just as before, Precede next applies to the $\langle [\text{PERS/NUM}], [\text{PERS/NUM}] \rangle$ set to order the subject and the rest of the clause. However, this time the two sets bear the same Pers/Num label, hence applying Precede yields the precedence relation of $\text{Pers/Num} \prec \text{Pers/Num}$, added to $\text{LP}(K)$ (15b). The rest of the structure undergoes the first step of Linearization as in (9) above (15c)–(15e).

(15) Precede yielding the set of precedence relations $\text{LP}(K)$ for (13)

- a. $\text{Precede}(C_{[Q]}, \langle [\text{PERS/NUM}], [\text{PERS/NUM}] \rangle) = C_{[Q]} \prec \langle [\text{PERS/NUM}], [\text{PERS/NUM}] \rangle \text{ LP}(K)$
 $= \{C_{[Q]} \prec \langle [\text{PERS/NUM}], [\text{PERS/NUM}] \rangle\}$ [by head/set]
- b. $\text{Precede}(\text{Pers/Num}, \text{Pers/Num}) = \text{Pers/Num} \prec \text{Pers/Num}$ [by goal/probe]
 $\text{LP}(K) = \{C_{[Q]} \prec \langle [\text{PERS/NUM}], [\text{PERS/NUM}] \rangle, \text{Pers/Num} \prec \text{Pers/Num}\}$
- c. $\text{Precede}(\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [Q]}, \text{Asp}) = \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [Q]} \prec \text{Asp} \text{ LP}(K)$
 $= \{C_{[Q]} \prec \langle [\text{PERS/NUM}], [\text{PERS/NUM}] \rangle, \text{Pers/Num} \prec T, T_{[\text{TENSE}], [\text{PERS/NUM}], [Q]} \prec \text{Asp}\}$ [by head/set]
- d. $\text{Precede}(\text{Pers/Num}_{[Q]}, n) = \text{Pers/Num}_{[Q]} \prec n$ [by head/set]
 $\text{LP}(K) = \{C_{[Q]} \prec \langle [\text{PERS/NUM}], [\text{PERS/NUM}] \rangle, \text{Pers/Num} \prec \text{Pers/Num}, \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [Q]} \prec \text{Asp}, \text{Pers/Num}_{[Q]} \prec n\}$

As before, Linearize creates the subject set $O(K)$ containing $\text{Pers/Num}_{[Q]} \cap n_{[Q]}$ and the set $L(K)$ ordering the verbal terminals (16a).

(16) Ordering statements for terminal nodes in (13)

$$O(K) = \{\text{Pers/Num}_{[Q]} \cap n_{[Q]}\} \quad L(K) = \{\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [Q]} \cap \textit{smiling}\}$$

The crucial difference between the English example in (7) and the pro-drop pattern in (12) comes into play at the point when an attempt is made at translating the relation $\text{Pers/Num} \prec \text{Pers/Num}$ included in $\text{LP}(K)$ in step (15b) into an ordering statement concatenating $O(K)$ and $L(K)$. Namely, with the two Pers/Num labels being indistinguishable in $\text{LP}(K)$, an ordering statement cannot be unambiguously determined for the terminals included in the structure labeled by the subject and

the clausal Pers/Num heads. This means that there is no way to integrate the two sets of ordering statements.¹⁸ I propose that to salvage the derivation and make externalization possible one of them needs to be deleted and, as is clear from the empirical facts, it is the subject set (17a). The motivation for the selection of the subject set for deletion remains to be investigated, though it can be speculated to follow from principles such as deletion up to recoverability (see section 4.4 below).¹⁹ The derivation continues by adding $C_{[Q]}$ to $L(K)$, completing Linearization of the clause (17b).

(17) Ordering statements for terminal nodes in (13) following rescue by deletion

- a. $O(K) = \{ \text{Pers/Num}_{[Q]} \frown n_{[Q]} \}$ [subject, deletion to enable externalization]
 $L(K) = \{ \text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]} \frown \textit{smiling} \}$ [clause]
- b. $L(K) = C_{[Q]} \frown \text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]} \frown \textit{smiling}$
 $\emptyset \quad \textit{is} \quad \textit{smiling}$ [clause]

To reiterate, on the current account the crucial difference between non-pro-drop and pro-drop languages lies in the labeling of the traditional T/T' as either T or Pers/Num and the consequences this has for Linearization. While in non-pro-drop languages no issues arise and nothing is deleted, since there is no trigger for deletion, the labels of the subject and its sister being different (see (7)), ambiguity arising in pro-drop languages due to label identity (both the subject and its sister are labeled as Pers/Num; see (13)) needs to be resolved, which—I propose—triggers subject deletion. With this discussion of the core of the proposal in place, I now proceed to supporting it further with the K/P/S data, incorporating the phenomenon of agreement drop into the picture.

3 Null subjects and null agreement in K, P, and S

On a superficial level, K, P, and S are morphosyntactically alike: all three languages have rich morphology, obligatorily inflect the verb for the person and number features of the subject in the present and future tenses (see Table 1 for the present tense;²⁰ the future tense paradigm is the same, with the inflections expressed either on the auxiliary (future imperfective) or directly on the lexical verb (future perfective)), and allow pro-drop. The naturally-occurring corpus examples in (18)–(20) illustrate this for the 2nd person singular.²¹

¹⁸The issue does not arise with lexical NPs, where the labels of the subject and its sister are different, since in lexical NPs number is not bundled with person (e.g., [Num [n [√]]] in D-less languages (see footnote 3)), unlike in pronouns (Ghomeshi and Massam 2020).

¹⁹Deletion of the ordering statements for subject terminals ($O(K)$) can be taken to trigger deletion of the associated precedence relations in $LP(K)$, and the subject representation in the hierarchical PF structure of the clause on the basis of which $LP(K)$ has been constructed, if still accessible at this point in the derivation.

²⁰Thanks for the K and S data in Tables 1–3 go to Marika Jocz and Bartłomiej Wanot.

²¹In this article, all generalizations about K, P, and S have been either based on corpus data, or checked with native speaker consultants, or confirmed through an experiment involving naive native speakers (see the Appendix for details).

	Kashubian	Polish	Silesian
1SG	czëtó-m	czyta-m	czytũ-m
2SG	czëtô-sz	czyta-sz	czyto-sz
3SG	czëtô-ø	czyta-ø	czyto-ø
1PL	czëtó-më	czyta-my	czytũ-my
2PL	czëtô-ta	czyta-cie	czyto-će
3PL	czëta-ją	czyta-ją	czyta-jům

Table 1: Present tense for ‘to read’

- (18) Mò-sz prôwdã [...]
have-2SG rightness
‘You_{SG} are right [...]’ [K, corpus file k7.txt in Ruda et al. 2022a]
- (19) Może ma-sz rację.
perhaps have-2SG rightness
‘Perhaps you_{SG} are right.’ [P, National Corpus of Polish, Przepiórkowski et al. 2012]
- (20) Mõ-sz ôdy mie nōmer, pra?
have-2SG from me number right
‘You_{SG} ’ve got my number, right?’ [S, corpus file s19.txt in Ruda et al. 2022b]

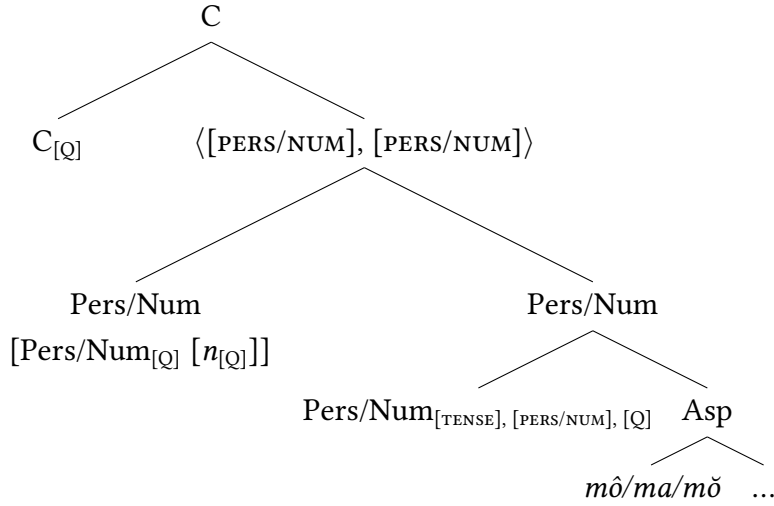
The derivation of examples of this type proceeds along the lines of (12)–(17) above, with a difference being that in the present tense in K/P/S agreement is realized as a suffix on the finite verb.²² Within the current assumptions this can be modeled as taking Precede(Pers/Num_{[TENSE], [PERS/NUM], [Q]}, Asp) from (15c) above to be blocked by lexical specification of the suffixal Pers/Num_{[TENSE], [PERS/NUM], [Q]}, with Asp_[Q] $\prec$ Pers/Num_{[TENSE], [PERS/NUM], [Q]} added to the set of precedence relations LP(K) instead of Pers/Num_{[TENSE], [PERS/NUM], [Q]} $\prec$ Asp (see footnote 12). However, as above, the label identity of the subject and its sister rendering Pers/Num $\prec$ Pers/Num in LP(K) requires subject deletion for final integration of the ordering statements to be possible. I show this in an abbreviated format in (21).

Glosses follow Leipzig conventions: 1, 2, 3: 1st, 2nd, and 3rd persons; INSTR: instrumental case; PL: plural; PRT: discourse particle; SE: reflexive; SG: singular. In addition, *l* indicates the *l*-participle form of a verb; M1 is the masculine personal gender, distinguished morphologically from the rest in the plural.

²²As noted above, the same inflection is present in future structures, whose pro-drop derivation parallels (21), with either the lexical verb or the auxiliary bearing suffixal agreement.

(21) Simplified derivation for (18)–(20)

a.



- b. $LP(K) = \{C_{[Q]} \prec \langle [PERS/NUM], [PERS/NUM] \rangle, Pers/Num \prec Pers/Num, m\hat{o}/ma/m\check{o} \prec Pers/Num_{[TENSE], [PERS/NUM], [Q]}, Pers/Num_{[Q]} \prec n\}$
- c. $\Theta(K) = \{Pers/Num_{[Q]} \frown n_{[Q]}\}$ [subject, deletion to enable externalization]
 $L(K) = m\hat{o}/ma/m\check{o} \frown Pers/Num_{[TENSE], [PERS/NUM], [Q]}$ [clause]
 $m\hat{o}/ma/m\check{o} \quad -sz$

However, an investigation of the past tense in K, P, and S reveals microvariation. In the past tense, verbs in these languages appear in a special participle form that inflects for number, gender, but not person—the so-called *l*-participle form. As shown in (22), the Polish *l*-participle co-occurs with an agreement clitic auxiliary, which reflects inflection for person and number.²³

(22) Polish past tense *l*-participle and [PERS/NUM] agreement auxiliary (bolded)

- | | | | |
|----|------------------------------------|----|--|
| a. | czytała- m | b. | czytała- ś |
| | read.L.SG.F-1SG | | read.L.SG.F-2SG |
| | ‘I read’ [singular female speaker] | | ‘you _{SG} read’ [singular female addressee] |

Importantly, K, P, and S vary in whether the [PERS/NUM] agreement auxiliary for the 1st and 2nd person must be realized overtly, as I will describe in more detail shortly. Based on the available descriptions, fieldwork with native speaker consultants, and experimental judgment surveys with naive native speakers (see the Appendix), the following generalization can be observed to hold across K, P, and S: when a [PERS/NUM] agreement auxiliary for the 1st and 2nd person is omitted (if omission is possible), pro-drop is blocked.

²³An argument coming from stress placement suggests that for some P speakers the agreement marker is a suffix rather than a clitic (or rather the two realizations co-exist in such speakers; see Franks and Bański 1999).

3.1 Microvariation in 1st and 2nd person auxiliaries and pro-drop

As a baseline, consider P: *l*-participle verbs must co-occur with the [PERS/NUM] agreement auxiliary in the 1st and 2nd persons. In (Central) K, however, no such overt auxiliary is available (Table 2).²⁴ Crucially, this gap in the auxiliaries corresponds with obligatory overt 1st and 2nd person subjects with *l*-participle verbs in K. No such restriction exists for P, where the auxiliaries are obligatory and pro-drop is possible.

subject	K overt subject & no auxiliary		P (null) subject & overt auxiliary	
1SG	*(jô) czëta(ła)	‘I read.L.SG.F’	(ja) czytała*(-m)	‘I read.L.SG.F-1SG’
2SG	*(të) czëta(ła)	‘you.SG read.L.SG.F’	(ty) czytała*(-ś)	‘you.SG read.L.SG.F-2SG’
1PL	*(më) czëtałë	‘we read.L.PL.F’	(my) czytały*(-śmy)	‘we read.L.PL.F-1PL’
2PL	*(wa) czëtałë	‘you.PL read.L.PL.F’	(wy) czytały*(-ście)	‘you.PL read.L.PL.F-2PL’

NB. (Central) K lacks the equivalents of the P [PERS/NUM] agreement auxiliaries *-m*, *-ś*, etc.

Table 2: Kashubian and Polish past tense paradigm for *l*-participle form of ‘read’

S diverges from K and P in small but important ways. Like P and unlike K, S has overt 1st and 2nd person [PERS/NUM] agreement clitic auxiliaries. The auxiliary is obligatory for 2nd person subjects. However, unlike in P, the auxiliary is optional in S for 1st person subjects (Table 3). Crucially, for 1st person subjects, pro-drop is only possible when the auxiliary is overtly expressed; 1st person subjects must be overtly expressed when the auxiliary is missing, like in K.

subject	overt subject & no auxiliary	(null) subject & overt auxiliary	
1SG	*(jô) czytała	(jô) czytała-’ch	‘I read.L.SG.F-1SG’
2SG	[auxiliary always overt]	(ty) czytała-’ś	‘you.SG read.L.SG.F-2SG’
1PL	*(my) czytały	(my) czytały-’chmy	‘we read.L.PL.F-1PL’
2PL	[auxiliary always overt]	(wy) czytały-’ście	‘you.PL read.L.PL.F-2PL’

Table 3: Silesian past tense paradigm for *l*-participle form of ‘read’

Note that no amount of contextual support can save pro-drop in the absence of the auxiliary in K and S. For example, even though the intended 1st person interpretation is obvious in the S example in (23), the auxiliary on the verb still must be overt for a null subject to be possible: combining pro-drop with the non-realization of [PERS/NUM] agreement is unacceptable here (see also Toman 1980; Veselovská 2020 for a similar observation for Czech, where both pro-drop and 1st person auxiliary omission are possible, but not at the same time).²⁵

²⁴Note that other K dialects manifest other patterns for the past tense (see Lorentz 1971; Breza 2001; Treder 2006; Jocz forthcoming).

²⁵I am grateful to Grzegorz Kulik for a detailed discussion of these data.

- (23) A: Co-ś robiła wczorej na wieczór?
 what-2SG did.L.SG.F yesterday on evening
 ‘What did you do yesterday evening?’
- B: *Suchała dobryj muzyki.
 listened.L.SG.F good music
Intended: ‘I listened to good music.’
The only possible reading: ‘She listened to good music.’
Cf. Suchała-ch dobryj muzyki ‘[I] listened.L.SG.F-1SG to good music.’ [S]

Following Ruda (2017), I assume that the omission of the 1st (and 2nd) person auxiliary constitutes the PF phenomenon of [PERS/NUM] agreement drop (see section 3.4 for a technical implementation).²⁶ The application of agreement drop, just like other PF-level rules, varies by language. In K, it applies obligatorily, but is restricted to 1st and 2nd person. In S, it is optional and restricted to 1st person only. However, in either case, when [PERS/NUM] agreement drop applies, pro-drop is blocked.

3.2 Third person auxiliaries and pro-drop

For thoroughness, I note that 3rd person subjects can undergo pro-drop even though there is no overt 3rd person [PERS/NUM] agreement auxiliary in K, P and S; see (24)–(26) for naturally-occurring corpus examples. This is not surprising in light of the fact that there is no overt 3rd person agreement marking for structures with *l*-participles more generally across Slavic languages. I thus suggest that agreement drop does not apply in the 3rd person. Rather, following a standard approach for Slavic (see, e.g., Franks 1995; Witkoś 1998; Franks and King 2000; Migdalski 2006), I assume that 3rd person agreement happens to be phonologically null ($-\emptyset$) in all three languages in the lexicon. That is, 3rd person auxiliaries are analyzed as null [PERS/NUM] agreement morphemes (e.g., K: (*òna*) *czëta(ła)*- $\emptyset$ /P: (*ona*) *czytała*- $\emptyset$ /S: (*uńna*) *czytała*- $\emptyset$ ‘she read.L.SG.F-3SG’), but the person and number features necessary for licensing a null subject are present in the clause both in the narrow syntax and at PF.

²⁶A more radical alternative would be to assume that the agreement features are not included in the auxiliaryless derivation at all. While this is also compatible with the approach to pro-drop suggested here, it would have more serious narrow syntactic ramifications (a.o., for nominative valuation), whose investigation is beyond the present scope, and it would raise questions about how to model the [PERS]-based restrictions on the auxiliary omission.

While I focus on the past tense here, I note that these patterns of subject and agreement drop extend to the present tense of the K and S copula, which requires (in Central K) or allows (in S) the non-realization of person agreement (e.g., *jô je* ‘I be.PRES.SG’ in K and *jo je* ‘I be.PRES.SG’ in S; cf. *jest-em* ‘be.PRES-1SG’ in P), as well as to the subjunctive and conditional moods.

	1SG/PL			2SG/PL			3SG/PL		
	P	K	S	P	K	S	P	K	S
Overt subject V + Aux _[PERS/NUM]	✓	*	✓	✓	*	✓	[PERS/NUM]		
<i>pro</i> V + Aux _[PERS/NUM]	✓	*	✓	✓	*	✓	auxiliary		
Overt subject V	*	✓	✓	*	✓	*	is [-ø]		
<i>pro</i> V	*	*	*	*	*	*			

Table 4: Microvariation in K/P/S subject and person/number agreement realization

- (24) Svòjã maszingerã szczerovól flot v mòjã stronã.
 self's submachine.gun directed.L.SG.M quickly in my direction
 'He quickly pointed his submachine gun at me.' [K, corpus file k78.txt in Ruda et al. 2022a]
- (25) Minął świeżo wykopane cztery groby [...]
 walked.past.L.SG.M freshly dug four graves
 'He walked past four freshly dug graves [...]'
 [P, National Corpus of Polish, Przepiórkowski et al. 2012]
- (26) Łoł woda, gasiół fojery, jeździół pōmogać [...]
 poured.L.SG.M water put.out.L.SG.M fires went.L.SG.M help
 'He poured water, put out fires, went to help [...]' [S, corpus file s73.txt in Ruda et al. 2022b]

All in all, the options of the realization of the person and number features of the subject are the same in the K, P and S past tense in the 3rd person, but they differ in the 1st and 2nd person, for which P and S but not (Central) K have overt [PERS/NUM] agreement auxiliaries (see Table 4). The auxiliaries can co-occur either with overt or null subject pronouns. In K the lack of [PERS/NUM] agreement marking enforces the use of overt subject pronouns in the 1st and 2nd person. This is also an option available to S in the 1st person, but not in the 2nd person, where S requires overt [PERS/NUM] agreement. A combination of no agreement and a null subject has not been attested for the 1st and 2nd person in any of the three languages.

3.3 Interim summary: a role for PF in licensing *pro*

The K, P, S microvariation described above suggest two ways for a [PERS/NUM] agreement auxiliary to go unrealized in the past tense, with different consequences for *pro*-drop. First, the auxiliary might be absent due to a PF deletion operation, as in the 1st(/2nd) person for K and S (see section 3.4 for an implementation). Alternatively, the auxiliary might be present, as a head with a full set of features, but it happens to be phonologically null, as in the 3rd person for K, P, and S. Importantly, this difference correlates with the availability of *pro*-drop: null subjects are

blocked in the absence of an overt 1st(/2nd) person auxiliary in K and S, but available in the 3rd person in all three languages.

This kind of sensitivity to how agreement is represented and realized is challenging for the accounts of consistent null subject languages which take pro-drop to be licensed entirely within the narrow syntax, independent of how agreement is ultimately realized at PF. At a minimum, I conclude from the data that PF licensing must not be overlooked in constructing theoretical approaches to pro-drop in rich agreement languages.

3.4 The K/P/S derivations

Recall first that the past tense in K, P, and S is formed with a *l*-participle and an auxiliary [PERS, NUM] agreement clitic (which is dropped under the conditions described above). Following the standard literature on Slavic morphosyntax (see esp. Borsley and Rivero 1994; Embick 1995; Szczegielniak 1995, 2005; Witkoś 1998; Franks and Bański 1999; Franks and King 2000; Migdalski 2006; Franks 2010), I assume that the [PERS(ON)] and [NUM(BER)] agreement features reflected on the auxiliary originate in the traditional T head, which on the current proposal is labeled as Pers/Num in pro-drop languages. The auxiliary is a clitic, autonomous from the verb, and it can either follow the *l*-participle, which I take to realize V-Voice-Asp (see Tables 2 and 3), or lean onto a preceding host, as shown in (27)–(28).²⁷

(27) No, dawno-**śmy** się nie widzieli.

PRT long.time-1PL SE not see.L.PL.M1

‘Right, we haven’t seen each other for a long time.’

[P, National Corpus of Polish, Przepiórkowski et al. 2012]

(28) Jo sama-**ch** siykła wszyjsko kosom [...]

I alone-1SG cut.L.SG.F everything scythe.INSTR

‘I cut everything with a scythe myself [...]’

[S, corpus file s14.txt in Ruda et al. 2022b]

I take the separate realization of [PERS/NUM] and [NUM/GEN] in verbal morphology at face value and assume that there are different [φ] probes in the clausal spine: a [PERS/NUM] probe in the clausal Pers/Num_{[TENSE], [PERS/NUM]} head and a [NUM/GEN] probe in Asp, which is realized as inflection of the *l*-participle.²⁸

²⁷I make these assumptions about auxiliary and verb placement here for explicitness, with alternative options like verb movement to Pers/Num (T/I; see Borsley and Rivero 1994) likewise being compatible with the analysis of pro-drop suggested here.

²⁸Importantly, there is no agreement morpheme in K, P, or S which encodes [PERS/NUM/GEN] simultaneously, despite the languages being highly fusional and encoding [PERS/NUM] and [NUM/GEN] as single morphemes. What is more, unlike the [PERS/NUM] probe, which is realized only once per clause, there can be multiple [NUM/GEN] probes, e.g., in past passive sentences, where both the *l*- and the *n/t*-participle inflect for [NUM/GEN] (e.g., *Kawa został-a wypit-a* ‘coffee_[F] became.L-SG.F drunk.N/T-SG.F’ (‘The coffee has been drunk’)). Finally, Citko (2018) shows that the

As above, I assume a pronominal representation of the subject following Ghomeshi and Mas-sam (2020), for whom [PERSON] is bundled with [NUMBER] in pronouns. I take [GENDER] to be contained separately in the nominalizing head *n* (Lowenstamm 2008; Willim 2012), the lowest functional head in the projection of the pronoun.²⁹ This yields the following basic representation of the clause in K, P, and S:

- (29) [Pers/Num_{[TENSE], [PERS/NUM]} [Asp_[NUM/GEN] [[Subject Pers/Num *n*P_[GEN]] Voice ...]]]

To explain the microvariation observed with respect to agreement realization, I propose that [PERS/NUM] agreement drop in K and S involves one more, intermediate step, that is Obliteration (terminal deletion) of Pers/Num_{[TENSE], [PERS/NUM]} before subject deletion takes place. The application of this rule varies by language, as mentioned above: in K, it applies obligatorily to past tense Pers/Num bearing 1st and 2nd person features, while in S, it applies only to past tense Pers/Num with 1st person features, and only optionally.³⁰ Jointly with the proposal for the derivation of pro-drop put forward above, these assumptions lead to the following derivations of null subjects, null agreement, and overt subjects in K, P, and S.

Null subjects As outlined in (14) above, on the current proposal a null subject is a consequence of label identity and the resultant unlinearizability of the terminals, resolved by subject deletion. In clauses like *Czytał-a-m/Czytał-a-'ch* ‘read.L-F.SG-1SG’ (‘I read’) in P/S, I take the verb to realize $\sqrt{[Q]}-v_{[Q]}-\text{Voice}_{[Q]}-\text{Asp}_{[Q]}$, $\text{Asp}_{[Q]}$ containing the [NUMBER] and [GENDER] agreement features, and I assume movement of the subject to the specifier of the clausal Pers/Num_{[TENSE], [PERS/NUM], [Q]} head, following Agree valuing the agreement features. This results in configuration (30a). (In what follows I omit C as irrelevant.) Chain Reduction and the Labeling Algorithm apply at PF, yielding (30b).

- (30) a. Subject movement (narrow syntax)

$$\overbrace{[[\text{Pers/Num}_{[Q]} [n_{[GEN], [Q]}]] [\text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]} [\text{Asp}_{[NUM/GEN], [Q]} [[\text{Pers/Num}_{[Q]} [n_{[GEN], [Q]}] [\textit{read} \dots]]]]]]}$$

[PERS/NUM] agreement (surfacing on the subjunctive complementizer in her examples) and the [NUM/GEN] inflection on the *l*-participle can be valued by different goals (bearing mismatching number features) when the subject is an &P. This provides further support for the analysis of structures with *l*-participles assuming separate [φ] probes in the clausal spine. See Veselovská (2020) for a similar conclusion, reached independently based on Czech data.

²⁹In Saab’s (2020; 2024) original analysis, the subject is a clitic with [PERSON], [NUMBER], and [GENDER] features bundled on a single head. This analysis is not adequate for K/P/S, which lack overt pronominal clitics (see, e.g., Cetnarowska 2004; Jung and Migdalski 2015; Migdalski 2016; Ruda 2021 for P), as well as the [GEN(ER)] feature in T (Pers/Num), as discussed above.

³⁰As 3rd person agreement marking is lexically null, there is no reason to postulate an additional Obliteration operation in this case, which is why Pers/Num is present in the clausal spine throughout the PF derivation, requiring 3rd person subject deletion (see, e.g., (24)–(26) above). Note that since Linearization applies before Vocabulary Insertion (Arregi and Nevins 2012), it targets terminals regardless of their ultimate phonemic realization (see Mendes et al. 2024 for a recent argument from ellipsis).

b. Chain Reduction and Labeling (PF)

$$\begin{aligned} & [\langle [\text{PERS/NUM}], [\text{PERS/NUM}] \rangle [\text{Pers/Num Pers/Num}_{[\text{PERS/NUM}], [\text{Q}]} [n_{[\text{GEN}], [\text{Q}]}]] [\text{Pers/Num Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}] \\ & [\text{Asp } \textit{read.L.F.SG} \dots]] \end{aligned}$$

When this structure is input to Precede, two outcomes are possible, reflecting the observation that the agreement auxiliary can either precede or follow the verb (see, e.g., the question and the appropriate version of the answer in (23) above for S). First (31a), $\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}$ can be ordered before the *l*-participle based on the default head/set setting, in which case the auxiliary attaches to any host that precedes it (see (27)–(28) above). Second (31b), the default ordering can be overridden, which in traditional terms could be taken to follow from auxiliary-to-participle head Lowering or participle-to-auxiliary head Raising. While nothing hinges on this here, I instead attribute this pattern to establishing the precedence relations during Linearization, with the option for the participle $\prec$ auxiliary ordering being lexically encoded in the auxiliary (see footnote 12). This derives the data without the need for an additional PF operation, whose possible application would likewise need to be encoded lexically.

(31) Basic intuition for the auxiliary/participle ordering

- a. Precede adds $\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]} \prec \text{Asp}$ (i.e., $\text{Aux}_{[\text{PERS/NUM}]} \prec$ participle set) to the set of precedence relations LP(K) by the head/set setting or
- b. Precede adds $\text{Asp}_{[\text{Q}]} \prec \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}$ (i.e., participle head $\prec \text{Aux}_{[\text{PERS/NUM}]}$) to the set of precedence relations LP(K) by lexical specification of $\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}$ (i.e., $\text{Aux}_{[\text{PERS/NUM}]}$).

Looking at (30) more closely, here, as in (15) above, Precede first yields the problematic output of the $\text{Pers/Num} \prec \text{Pers/Num}$ precedence relation (32a). At the next step, Precede applies to the clausal Pers/Num set (the X' -theoretic T'), where the agreement auxiliary and the *l*-participle originate. As outlined in (31), two scenarios are possible at this stage (32b). The subject-internal ordering is established next, or in a separate cycle (32c).

(32) Precede yielding the set of precedence relations LP(K) for (30b)

- a. $\text{Precede}(\text{Pers/Num}, \text{Pers/Num}) = \text{Pers/Num} \prec \text{Pers/Num}$ [by goal/probe]
 $\text{LP(K)} = \{\text{Pers/Num} \prec \text{Pers/Num}\}$
- b. $\text{Precede}(\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}, \text{Asp}) = \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]} \prec \text{Asp}$
 $\text{LP(K)} = \{\text{Pers/Num} \prec \text{Pers/Num}, \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]} \prec \text{Asp}\}$ [by head/set]
or $\text{Precede}(\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}, \textit{read.L.F.SG}) =$
 $\textit{read.L.F.SG} \prec \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}$ [by lexical specification]
 $\text{LP(K)} = \{\text{Pers/Num} \prec \text{Pers/Num}, \textit{read.L.F.SG} \prec \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}\}$
- c. $\text{Precede}(\text{Pers/Num}_{[\text{Q}]}, n) = \text{Pers/Num}_{[\text{Q}]} \prec n$ [by head/set]

$$\begin{aligned}
& \text{LP(K)} = \{\text{Pers/Num} \prec \text{Pers/Num}, \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]} \prec \text{Asp}, \text{Pers/Num}_{[\text{Q}]} \prec n\} \\
& \text{or} \quad \text{LP(K)} = \{\text{Pers/Num} \prec \text{Pers/Num}, \text{read.L.F.SG} \prec \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}, \text{Pers/Num}_{[\text{Q}]} \prec n\}
\end{aligned}$$

When these precedence relations are translated into the ordering statements for terminals, the ambiguity arising from $\text{Pers/Num} \prec \text{Pers/Num}$ triggers deletion of the subject ordering statements, along the lines of (17) above. The result are the ordering statements in (33), externalized as a null subject clause.

(33) Ordering statements for terminal nodes in (30b) following rescue by deletion

$$\begin{aligned}
& \Theta(\text{K}) = \text{Pers/Num}_{[\text{Q}]} \frown n_{[\text{Q}]} && [\text{subject}] \\
& \text{L(K)} = \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]} \frown \text{read.L.F.SG} \\
& \quad \text{Aux}_{[\text{PERS/NUM}]} \quad l\text{-participle} && [\text{clause}] \\
& \text{or} \quad \Theta(\text{K}) = \text{Pers/Num}_{[\text{Q}]} \frown n_{[\text{Q}]} && [\text{subject}] \\
& \text{L(K)} = \text{read.L.F.SG} \frown \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]} \\
& \quad l\text{-participle} \quad \text{Aux}_{[\text{PERS/NUM}]} && [\text{clause}]
\end{aligned}$$

Null agreement This occurs in clauses like *Źo czyt-a/Źo czyt-a* ‘I read.L-F.SG’ (‘I read’) in K/S when the clausal $\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}$ is obliterated prior to Linearization, in which case the Linearization ambiguity does not arise. In particular, following the same narrow syntactic derivation as in (30a), Obliteration applies at PF, in addition to Chain Reduction, removing $\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}$ (34a).³¹

In this case, after the application of the Labeling Algorithm (34b), the labels which Precede identifies at the top are the subject Pers/Num and Asp , allowing for an unambiguous relation of $\text{Pers/Num} \prec \text{Asp}$ to be added to LP(K) by reference to the goal/probe relation (Asp and the subject Pers/Num have been linked by Agree valuing the $[\text{NUM/GEN}]$ probe on Asp) (35), resulting in unproblematic establishment of the ordering statements in (36) and deriving an overt subject with agreement drop.

(34) a. Obliteration and Chain Reduction at PF

$$[[\text{Pers/Num}_{[\text{Q}]} [n_{[\text{GEN}], [\text{Q}]}]] [\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]} [\text{read.L.F.SG} [[\text{Pers/Num}_{[\text{Q}]} [n_{[\text{GEN}], [\text{Q}]}]] \dots]]]]$$

b. Labeling (PF)

$$\langle [(\text{NUM}), (\text{NUM})] \rangle [\text{Pers/Num} \text{ Pers/Num}_{[\text{PERS/NUM}], [\text{Q}]} [n_{[\text{GEN}], [\text{Q}]}]] [\text{Asp}_{[\text{NUM/GEN}]} \text{read.L.F.SG} \dots]]$$

³¹While seeking a deeper explanation of the application and restrictions on Obliteration would be a natural step forward, for the present purpose it is sufficient to stipulate the condition on Obliteration in K and S as $_\text{[/PAST, PERS:1(2)]}$, which also captures the fact that the auxiliary ‘be’ in the future tense is not dropped.

- (35) Precede yielding the set of precedence relations LP(K) for (34b)
- a. Precede(Pers/Num, Asp) = Pers/Num $\prec$ Asp [by goal/probe]
LP(K) = {Pers/Num $\prec$ Asp}
- b. Precede(Pers/Num_[Q], *n*) = Pers/Num_[Q] $\prec$ *n* [by head/set]
LP(K) = {Pers/Num $\prec$ Asp, Pers/Num_[Q] $\prec$ *n*}
- (36) Ordering statements for terminal nodes in (34b)
- Pers/Num_{[PERS/NUM], [Q]} $\frown$ *n*_{[GEN], [Q]} $\frown$ *read.L.F.SG*
subject pronoun *l*-participle

- (37) a. Subject movement (narrow syntax)
-
- $$[\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [Q]} \quad [[\text{Pers/Num}_{[Q]} [n_{[\text{GEN}], [Q]}]] \quad [\text{Asp}_{[\text{NUM/GEN}], [Q]} \quad [[\text{Pers/Num}_{[Q]} [n_{[\text{GEN}], [Q]}] \dots]]]]$$
- b. Labeling (PF)
- $$[\text{Pers/Num} \quad \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [Q]} \quad [\langle \text{Num}, \text{Num} \rangle \quad [\text{Pers/Num} \quad \text{Pers/Num}_{[\text{PERS/NUM}], [Q]} \quad [n_{[\text{GEN}], [Q]}]] \quad [\text{Asp}_{[\text{NUM/GEN}]} \quad \text{read.L.F.SG} \dots]]]$$

³³I abstract away from the Spec-to-Spec anti-locality issue which this derivation raises (see Branen 2023 for a recent discussion).

(38) Precede yielding the set of precedence relations LP(K) for (37b)

- a. $\text{Precede}(\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}, \langle \text{Num}, \text{Num} \rangle) = \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]} \prec \langle \text{Num}, \text{Num} \rangle$ [by head/set]
 $\text{LP}(K) = \{\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]} \prec \langle \text{Num}, \text{Num} \rangle\}$
or $\text{Precede}(\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}, \text{read.L.F.SG}) =$
 $\text{read.L.F.SG} \prec \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}$ [by lexical specification]
 $\text{LP}(K) = \{\text{read.L.F.SG} \prec \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}\}$
- b. $\text{Precede}(\text{Pers/Num}, \text{Asp}) = \text{Pers/Num} \prec \text{Asp}$ [by goal/probe]
 $\text{LP}(K) = \{\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]} \prec \langle \text{Num}, \text{Num} \rangle, \text{Pers/Num} \prec \text{Asp}\}$
or $\text{LP}(K) = \{\text{read.L.F.SG} \prec \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}, \text{Pers/Num} \prec \text{Asp}\}$
- c. $\text{Precede}(\text{Pers/Num}_{[\text{Q}]}, n) = \text{Pers/Num}_{[\text{Q}]} \prec n$ [by head/set]
 $\text{LP}(K) = \{\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]} \prec \langle \text{Num}, \text{Num} \rangle, \text{Pers/Num} \prec \text{Asp}, \text{Pers/Num}_{[\text{Q}]} \prec n\}$
or $\text{LP}(K) = \{\text{read.L.F.SG} \prec \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}, \text{Pers/Num} \prec \text{Asp}, \text{Pers/Num}_{[\text{Q}]} \prec n\}$

(39) Ordering statements for terminal nodes in (37b)

- a. $\text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]} \frown \text{Pers/Num}_{[\text{PERS/NUM}], [\text{Q}]} \frown n_{[\text{GEN}], [\text{Q}]} \frown \text{read.L.F.SG}$
 $\text{Aux}_{[\text{PERS/NUM}]}$ subject pronoun *l*-participle
- b. $\text{Pers/Num}_{[\text{PERS/NUM}], [\text{Q}]} \frown n_{[\text{GEN}], [\text{Q}]} \frown \text{read.L.F.SG} \frown \text{Pers/Num}_{[\text{TENSE}], [\text{PERS/NUM}], [\text{Q}]}$
subject pronoun *l*-participle $\text{Aux}_{[\text{PERS/NUM}]}$

For postverbal subjects, both pronominal and lexical ones, which are likewise available in K, P, and S (40), as is the case more broadly in consistent null subject languages, I suggest that they remain in Spec,Voice (in the contexts that do not warrant assumptions about further subject and/or verb movements).

- (40) a. [...] a leżała ona między Jōnkowicami a Markłowicami [...]
and lay.L.SG.F she between Jōnkowice and Marklowice
‘[...] and it was situated between Jōnkowice and Marklowice [...]’
[S, corpus file s62.txt in Ruda et al. 2022b]
- b. Cołki życi przoł on rachōnkōm [...]
whole life love.L.SG.M he maths
‘He loved maths his whole life [...]’
[S, corpus file s73.txt in Ruda et al. 2022b]

This does not result in a labeling issue under the assumption that the verb moves to $\text{Asp}_{[\text{Q}]}$ and its copy in $\text{Voice}_{[\text{Q}]}$ has been removed by the time the Labeling Algorithm applies, allowing for

the subject to provide the label for the structure (41).³⁴ Following Precede applied to the clausal Pers/Num_{[TENSE], [PERS/NUM], [Q]} head and Asp (42a), Precede then applies to the verb and the Pers/Num set labeled by the subject Pers/Num_[Q] head. By the head/set rule, the set containing the subject is ordered after the verb (42b).

(41) a. Subject *in situ*, V in Asp, Chain Reduction (PF)

$$\overbrace{[\text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]} [\text{lay.L.F.SG} [[\text{Pers/Num}_{[Q]} [n_{[GEN], [Q]}]] [\text{lay} \dots]]]]}$$

b. Labeling (PF)

$$[\text{Pers/Num} \text{ Pers/Num}_{[TENSE], [PERS/NUM], [Q]} [\text{Asp} \text{ lay.L.F.SG} [\text{Pers/Num} [\text{Pers/Num} \text{ Pers/Num}_{[PERS/NUM], [Q]} [n_{[GEN], [Q]}] \dots]]]]$$

(42) Precede yielding the set of precedence relations LP(K) for (41b)

a. $\text{Precede}(\text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]}, \text{Asp}) = \text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]} \prec \text{Asp LP(K)}$
 $= \{\text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]} \prec \text{Asp}\}$ [by head/set]

or $\text{Precede}(\text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]}, \text{lay.L.F.SG}) =$
 $\text{lay.L.F.SG} \prec \text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]}$ [by lexical specification]
 $\text{LP(K)} = \{\text{lay.L.F.SG} \prec \text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]}\}$

b. $\text{Precede}(\text{lay.L.F.SG}, \text{Pers/Num}) = \text{lay.L.F.SG} \prec \text{Pers/Num}$ [by head/set]
 $\text{LP(K)} = \{\text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]} \prec \text{Asp}, \text{lay.L.F.SG} \prec \text{Pers/Num}\}$

or $\text{LP(K)} = \{\text{lay.L.F.SG} \prec \text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]}, \text{lay.L.F.SG} \prec \text{Pers/Num}\}$

c. $\text{Precede}(\text{Pers/Num}_{[Q]}, n) = \text{Pers/Num}_{[Q]} \prec n$ [by head/set]

$\text{LP(K)} = \{\text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]} \prec \text{Asp}, \text{lay.L.F.SG} \prec \text{Pers/Num}, \text{Pers/Num}_{[Q]} \prec n\}$

or $\text{LP(K)} = \{\text{lay.L.F.SG} \prec \text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]}, \text{lay.L.F.SG} \prec \text{Pers/Num}, \text{Pers/Num}_{[Q]} \prec n\}$

The ordering statements for the terminals in (43) are the translations of the relations created in (42), as per (6) above. Both of these orderings are available in P and S, even if their naturalness depends on pragmatic contextualization.

(43) Ordering statements for terminal nodes in (41b)

a. $\text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]} \frown \text{lay.L.F.SG} \frown \text{Pers/Num}_{[Q]} \frown n_{[GEN], [Q]}$
 $\text{Aux}_{[PERS/NUM]} \quad \quad \quad l\text{-participle} \quad \quad \text{subject pronoun}$

b. $\text{lay.L.F.SG} \frown \text{Pers/Num}_{[TENSE], [PERS/NUM], [Q]} \frown \text{Pers/Num}_{[Q]} \frown n_{[GEN], [Q]}$
 $l\text{-participle} \quad \text{Aux}_{[PERS/NUM]} \quad \quad \quad \text{subject pronoun}$

³⁴If the verb moves to Asp and its lower copies are removed and hence do not participate in labeling also in languages like English, the EPP effect cannot be derived from the hypothesized labeling issue with VoiceP, *contra* Chomsky (2013, 2015). The alternative likewise postulated by Chomsky and pointing to the inability of T to label the structure as the driving force for the movement may thus be needed to account for the EPP requirement, if it indeed warrants a narrow syntactic explanation (see, e.g., Landau 2007; Sigurðsson 2002; McFadden and Sundaresan 2018).

In sum, this section has introduced a number of analytical details aiming at accounting for the specific patterns of microvariation observed in K, P, and S by employing the independently postulated mechanisms of Obliteration, labeling, and Linearization, as well as the assumption that derivations illicit at the SM interface can be rescued by PF deletion. To recapitulate, the general logic of this approach is that if [PERS/NUM] agreement features label the head that introduces them, subsequent (I-)Merge of the subject pronoun creates a configuration which cannot be linearized. This in turn triggers deletion of the subject pronoun, which resolves the problem and allows for the structure to be externalized.

4 Further theoretical issues

This section presents relatively tentative suggestions concerning the correlation between pro-drop and T bimorphemicity, a possible extension of the proposal to partial and radical pro-drop languages, as well as the question of the conditions on PF deletion.

4.1 *The bimorphemic T/pro-drop correlation*

Saab (2020, 2024) and Koenenman and Zeijlstra (under revision) independently propose that the key factor in licensing the null subject in consistent null subject languages is that agreement is expressed by a morpheme separate from the one expressing tense, that is the categories of tense and agreement are realized bimorphemically. This is achieved by agreement dissociation at PF in Saab (2020, 2024) and by postulating separate Agr and T heads in the narrow syntax in Koenenman and Zeijlstra (under revision). Implementational details aside, both approaches block null subjects in languages where [TENSE] and agreement are expressed with a single morpheme, even if the language has (relatively) rich agreement (e.g., Icelandic, which is not a pro-drop language). The expression of tense and agreement in K, P, and S is indeed bimorphemic too (see, e.g., Schenker 1954 for P³⁵). Pending more in-depth inquiry into the cross-linguistic applicability of the correlation and the possible representational and derivational mechanisms responsible for it, I offer the following suggestion for capturing the correlation under the Linearization-based approach to pro-drop.

In section 2.1.1, example (11), I have proposed the following distinction between pro-drop and non-pro-drop languages:

³⁵Adding to Schenker’s (1954) arguments, I note that the independence of agreement marking from tense in P/S is suggested by the [PERS/NUM] marking of present tense ‘be’, which is achieved by the same auxiliary clitics that accompany the *l*-participle in the past tense (e.g., for ‘you are’ and ‘you were’ in P: *jest-eś* ‘be.PRES-2SG’, *była-ś* ‘be.L.F.SG-2SG’ with clitic attachment to V, or, e.g., *że-ś jest* ‘that-2SG be.PRES’, *że-ś była* ‘that-2SG be.L.F.SG’ with pre-verbal clitic attachment).

(44) [TENSE] and [PERS/NUM] combinations into a head

a. ‘Poor’ agreement/‘weak’ $T \rightarrow \text{non-pro-drop}$

$T_{\{[TENSE], [PERS/NUM], [Q]\}} \rightarrow \text{Labeling Algorithm ‘sees’ } T$

b. ‘Rich’ agreement/‘strong’ $T \rightarrow \text{consistent pro-drop}$

$\text{Pers/Num}_{\{[TENSE], [PERS/NUM], [Q]\}} \rightarrow \text{Labeling Algorithm ‘sees’ Pers/Num}$

If heads are feature sets, monomorphemicity and bimorphemicity can be understood as a consequence of the feature set structures. In particular, I tentatively suggest that each subset constituting a head is associated with a morpheme. Whereas the feature set $\{[F_X], [F_Y]\}$ is externalized with a single morpheme, the set $\{[F_X], \{[F_Y], [Q]\}\}$ is externalized with two. If Vocabulary Insertion is the replacement of a Q-variable in a head by a morphological exponent (see footnote 7), this can be formalized further by taking each subset constituting a head to contain a Q-variable by default, i.e., $\{[F_X], [F_Y], [Q]\}$ vs. $\{[F_X], \{[F_Y], [Q]\}, [Q]\}$. If so, $T_{\{[TENSE], [PERS/NUM]\}}$ can be represented as $\{[TENSE], [PERS/NUM], [Q]\}$, whereas $\text{Pers/Num}_{\{[TENSE], [PERS/NUM]\}}$ can be represented as $\{ \{[TENSE], [Q]\}, [PERS/NUM], [Q] \}$, realized as two morphemes. While I put this idea forward as a future research direction here, I note that the current analysis of pro-drop is also compatible with agreement dissociation as suggested in Saab (2020, 2024), with fission of $\text{Pers/Num}_{\{[TENSE], [PERS/NUM]\}}$, as well as with separate Agr *qua* Pers/Num and T heads along the lines of Koenenman and Zeijlstra (under revision).

4.2 Expected patterns of variation: partial pro-drop

The empirical focus of the analysis presented above has been on West Slavic languages. Since pro-drop in South Slavic languages parallels Polish (see, e.g., Franks 1995), the same derivations can be assumed there. The situation seems more complex in East Slavic languages. For example, Franks (1995) describes Russian as a non-pro-drop language (Franks 1995:299) or a “mixed” null subject language (Franks 1995:300), ultimately concluding that it may best be described as a discourse pro-drop language akin to Chinese (Franks 1995:303). More recent studies classify Russian as a partial pro-drop system (e.g., Holmberg 2010; Barbosa 2019; Roberts 2019). However, this characterisation is questioned by Tsedryk (2015, 2022), who points out that Russian, as well as Belarusian and Ukrainian (also East Slavic languages), differ from non-pro-drop and partial pro-drop languages in that when the context is right, they allow null subjects without imposing any person- or tense-based restrictions. Echoing Franks, though drawing different analytical conclusions, Tsedryk argues that the restrictions on pro-drop in East Slavic are purely information structural and that null subjects in these languages are contextually-interpreted ϕ Ps (Tsedryk 2022:45), with overt pronouns being represented as DPs. This approach can easily be married with the current proposal for deriving the nullness of the subject at PF, with Tsedryk’s ϕ P being a Pers/Num projection, deleted along the lines of (14) above (see also Gribanova 2013, who

explicitly proposes that null subjects in Russian should be analysed as *pro* falling within Rizzi's 1986 licensing framework).³⁶ An important twist here is that, unlike West and South Slavic, East Slavic languages lack overt person marking in the past tense, which is similar to Central K (see, e.g., Table 2 above), but, unlike for K, this seems not to matter for the distribution of null subjects in East Slavic (for a discussion of Russian, see also, e.g., Madariaga 2018 and references therein).³⁷

The current proposal can account for the nullness of East Slavic subjects in the past tense on the assumption that T bears (zero-realized) [PERS/NUM] in this context nevertheless throughout the derivation, and no Impoverishment or Obliteration operations apply (at least prior to Linearization), unlike in K (cf. also Müller 2006). Alternatively, if the past tense T lacks agreement features in East Slavic altogether, the movement of a subject Pers/Num to Spec,T renders the resultant set (i.e., traditional TP) unlabelable (as the [XP YP] configuration), a situation that can be resolved by subject deletion at PF, along the lines suggested in section 4.3 below for argument ellipsis.³⁸ In this case null subjects in (the past tense in) East Slavic would indeed be derived in parallel to null arguments in discourse pro-drop languages, aligning with Franks's (1995) original intuition (cf. also the suggestions for unifying partial and radical pro-drop in Barbosa 2019 and Roberts 2019). Needless to say, establishing the adequacy of these suggestions or, if needed, developing further alternatives conceivable within the current framework, requires in-depth research focused on East Slavic.³⁹

Further patterns interesting from the perspective of the current analysis come, for example, from Continental West Germanic dialects such as (varieties of) Bavarian and Frisian, which typically restrict pro-drop to [2SG] subjects. Weiss (2005) argues that pro-drop in these dialects is possible only when the agreement marker expressed on the complementizer and/or the verb has originated by way of reanalysis of a subject clitic as inflection (see also Weiss and Volodina 2018). This has not been the case for the other [PERS/NUM] combinations, which do not license pro-drop

³⁶Tsedryk (2022) achieves the nullness of the subject by postulating language-specific spell-out rules for subject pronouns (e.g., [_{φP} φ [_{NP} n]] ↔ Ø (Tsedryk 2022:67)).

³⁷Just like West and South Slavic languages, East Slavic languages use *l*-participles in the past tense, marked for [NUM/GEN], introduced into the derivation on Asp on the current analysis. The present and future tense forms unambiguously express all [PERS/NUM] combinations.

³⁸Such a difference in the feature content of past tense T (clausal Pers/Num) between East Slavic (no φ) and Central K (obliterated φ) may not be groundless. Unlike speakers of East Slavic languages, K speakers are also typically native speakers of P, and are hence bilingual between two languages with quite similar morphosyntax. A hypothesis worth exploring could thus be that K speakers use the same clausal Pers/Num head in both languages, which therefore needs to include [PERS/NUM] regardless of tense, because P inflects for these features in all tenses. This is in line with López's (2020) approach to the bilingual lexicon/I-language as an integrated system.

³⁹This certainly includes research on the interpretive differences between null and overt pronouns across Slavic (and beyond; see, e.g., Frascarelli and Jiménez-Fernández 2019; Ruda and Huang forthcoming and references therein). I do not address this topic here, because these differences are information structural and pragmatic in nature and I subscribe to Šimik's (2021) approach postulating an indirect relation between information structure and syntax/semantics. On this view, "information structure is part of discourse pragmatics, which in turn is part of how language is used – so-called linguistic PERFORMANCE, while syntax and semantics constitute a system that is largely independent of language use and is part of what is often called linguistic COMPETENCE" (Šimik 2021:4). The current article is intended as a contribution to syntactic theory.

(with exceptions like Central Bavarian, where also [2PL] and [1PL] inflections are of pronominal origin and where null subjects can be not only [2SG], but also [1/2PL] (Weiss and Volodina 2018:272–273)). I suggest that the “pronominal” nature of [2SG] agreement (in Weiss’s terms) in Germanic dialects makes it more prominent for labeling than [TENSE] and hence the head containing [TENSE] and [2SG] is a [2SG] Pers/Num head, requiring deletion of the subject in its specifier for Linearization to succeed (see (14) above and related derivations). This could be due to the language acquirer detecting the morphological similarity between [2SG] agreement and pronouns and, as a result, setting for labeling the traditional [2SG] T and pronouns based on the same features (i.e., Pers/Num, regardless whether [2SG] is included in a pronominal or a clausal projection). On the current approach, partial pro-drop languages can thus be expected to be fundamentally mixed systems with the non-pro-drop, [TENSE]-based labeling overridden in some contexts by the consistent pro-drop, [PERS/NUM]-based labeling, when cued by morphological generalizations made by the acquirer.

Hypothetically, a partial pro-drop language could also go in the opposite direction and be mostly a clausal Pers/Num language, with this labeling type being unavailable in some contexts that would then be incompatible with null subjects. While more in-depth research is needed, this could perhaps be instantiated by Hebrew. In this language, in the present tense verbs inflect only for [NUM/GEN] and null subjects are blocked (Shlonsky 2009).⁴⁰ On the other hand, in the past and future tenses the verb inflects for [PERS/NUM] (as well as [GEN]) and null subjects are allowed. It is thus conceivable that while the agreement-based labeling is uniformly applied in Hebrew, there is no [PERS] in the present tense (by the time labeling applies), and the traditional T is therefore labeled as [NUM] (or [NUM/GEN] if in Hebrew, unlike in Slavic, [GEN] is included in the set introducing [TENSE]). If this is the case, the labels of the pronominal subject (i.e., Pers/Num) and the traditional T’ (i.e., Num(/Gen)) are distinct in the present tense and there is therefore no issue with establishing a linear order of the terminals and no trigger for subject deletion.

In sum, the current proposal offers two basic routes to partial pro-drop, which has the potential for accounting for the heterogeneity of this language type. Ultimately, all specific analytical choices in this space should be tied to morphosyntactic features that the acquirer can be expected to be able to detect in primary linguistic data.

4.3 *Labeling, rescue by deletion, and argument ellipsis*

Working within the assumptions of the anti-agreement approach, according to which only arguments which are not agreed with can be deleted (see Saito 2007, 2016; Takahashi 2014, 2020;

⁴⁰This holds unless the [PERS/NUM]-agreeing negator *eyn* is present, in which case null [1/2] subjects are licensed (see Shlonsky 2009). Third person subjects are constrained by interpretive restrictions (Shlonsky 2009) that in discussions of partial pro-drop have been captured with mechanisms such as null NP anaphora (Barbosa 2013, 2019) or control (Holmberg 2005, 2010; Holmberg et al. 2009; Holmberg and Sheehan 2010; non-standard binding/Control for Shlonsky 2009).

Roberts 2019 and related), Saab (2020) assumes that overt nominal arguments must have a valued [Case] feature at PF, with arguments lacking it obligatorily remaining unpronounced. While Saab proposes to capture this by taking [Case]-less (K-less) arguments to undergo deletion in the narrow syntax, I link this proposal here to Saito's (2016) hypothesis that [Case] in languages like Japanese serves as an anti-labeling device, with the constituent marked for [Case], null arguments included, being invisible to the Labeling Algorithm (i.e., label [XP_[Case] YP] = Y (suffixal [Case] on XP)). Consequently, if an argument in this language type lacks [Case], it cannot be ignored for labeling, which I propose causes a labeling failure triggering rescue by deletion, applied to the argument at PF as a means to make labeling and, what follows, Linearization possible. Thus, while this suggestion follows Saito's (2016) insight into the role of suffixal [Case] for labeling, unlike Saito (2016) and similarly to Saab (2020) I take argument ellipsis to crucially implicate the lack of [Case] on the argument (see esp. Oku 1998; Saito 2007), deriving what Saab refers to as an ellipsis/[Case] corollary. Pending more in-depth investigation, I sketch a potential derivation along these lines below.

Consider (45), an argument ellipsis version of the non-pro-drop and consistent pro-drop examples in (7) and (12) in section 2.1.1 above, understood to belong to a hypothetical radical pro-drop language, where—for expository purposes—the null argument is represented as a minimal *nP*, that is $\{n\}$ (Tomioka 2003; Kura Fuji 2018; Barbosa 2019), though the analysis can equally well be applied to fully-fledged lexical arguments.

Assuming Chain Reduction of the lower copy of n , the structure is labeled up to T by the head/set rule, at which point labeling the structure further is impossible due to the lack of [Case] on n (44), as in (46).

To salvage the derivation, rescue by deletion applies, removing n from the structure and allowing for the labeling to be completed, as in (47), which constitutes legitimate input to Precede proceeding in the manner outlined in the previous sections.⁴¹

along the following lines: ‘deletion can apply to α in $\{\alpha, \beta\}$ only if α and β have an identical label and α is “recoverable” from a term of β .’ While this covers [PERS/NUM], one potential issue is with [GENDER], which numerous pro-drop languages, including Czech, Italian, Spanish, as well as K, P, and S, express morphologically at least on some pronouns, but not necessarily on (all) verb forms, which nevertheless allow pro-drop. Another problem, as noted by the Reviewer, is that this does not account for argument ellipsis in radical pro-drop languages, as approached in this paper. A modification of the Reviewer’s formalization which could resolve both issues could be as follows: ‘deletion can apply to α in an unlinearizable $\{\alpha, \beta\}$ only if the feature content of α ’s head is “recoverable” from a term of β .’ This captures consistent pro-drop contexts in that the subject’s [PERS/NUM] are included in $T \text{ qua Pers/Num}_{\{[TENSE], [PERS/NUM]\}}$, but [TENSE] is not included in the subject, and [GENDER], introduced lower in the structure of the pronoun, need not be recoverable from T. It can also capture argument ellipsis on the *n*-based approach (Tomioka 2003; Kurafuji 2018; Barbosa 2019) on the assumption that the only feature which *n* bears in the relevant languages is a categorial feature, which could be taken to be recoverable from verbal selectional restrictions.

An alternative to conditions of this type is to take deletion to apply freely to unlinearizable structures in principle, but with the selection of the set to which it applies being guided by economy and minimalization of information loss, both of which would favor deletion of the ‘smaller’ set. Either way, considering these alternatives, as well as some others, should ideally be done against the background of conditions governing rescue by deletion in other context for which it has been postulated and the ones where it would help but cannot apply, hopefully allowing for a uniform treatment in the future, determining what morphosyntactic information can be lost on the way to the SM system.

5 Conclusion

The study of microvariation in K, P, and S reported here, whose empirical results are summarized in Table 4 above, shows that if morphology makes it possible (i.e., the deletion of agreement does not affect the realization of the lexical verb itself), a rich-agreement language that is otherwise a consistent null subject language can manifest agreement drop rather than pro-drop. The fact that [PERS/NUM] agreement drop, which I argue is best analyzed as a PF phenomenon, bleeds pro-drop is difficult to accommodate under the accounts in which the licensing of *pro* is restricted to the narrow syntax module and is insensitive to how agreement is realized at PF (cf. Müller 2006). I interpret this fact as positive evidence for the role of PF in deriving null subjects. Adapting Saab’s (2020; 2024) PF-based head ellipsis approach to pro-drop as my starting point, I have further proposed that the bleeding effect is observed when PF deletion is preceded by Obliteration applying to $T \text{ qua Pers/Num}$, removing the structural conditioning for its application.

I have aimed to motivate subject PF deletion by providing a precise formalization of the rela-

tion between label ambiguity, unlinearizability, and deletion. Such an analysis aligns with recent proposals that link ellipsis to requirements for labeling and/or Linearization (e.g., Takita 2020). Thus, under the current approach pro-drop is not a special phenomenon, but—with more research into deletion—may prove to be an instance of a broader family of elliptical constructions manifesting rescue by PF deletion, applying to salvage a derivation that could not otherwise be externalized.

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Null subjects, labeling, and Linearization: Microvariation in subject and agreement drop Supplementary Appendix

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Experiments on agreement in K, P, S

K and S are both languages that are underdescribed relative to P.¹ Being minority varieties, K and S also show language-internal variation. To ensure a robust empirical basis for understanding the [PERS/NUM] agreement auxiliary clitics available, we conducted acceptability judgment experiments, using auditory materials, testing to what extent the lack of [PERS/NUM] agreement auxiliaries was possible and acceptable. More specifically, the goal here was to determine the range of variation across these three varieties, taking P, a typical consistent null subject language, as a baseline for comparison. These experiments complement data collected through fieldwork and corpora.

We constructed three combinations of null/overt subjects and present/absent [PERS/NUM] agreement auxiliaries, excluding a fourth combination without both subject and the agreement, because these sentences are in principle ambiguous: they have an alternative grammatical parse in which the subject is understood to be in the third person. One could have provided additional context in an effort to disambiguate the sentences, but doing so would have added to the complexity and length of the task, which we judged to be undesirable.

As our investigation of K was based on the Central dialect, which is said to lack overt [PERS/NUM] agreement markers in the past tense, the markers that were used in the experimental materials were added by a native speaker translator based on the markers available in other dialects (see footnote 4 in main paper).

Materials

For each language, twenty-four frames were constructed by including the tested structures in embedded clauses, with the matrix clause containing a proper name subject followed by the verbs “say” or “think”. Embedded subjects were evenly distributed over the [1/2] person, as well as [SG/PL] number combinations (four person-number combinations, six sentences per combination); doing so let us verify our observations about S (namely, first person agreement can be either overt or not, while second person agreement must be overt). Three variants were created for each sentence, one per each of the three subject/agreement combinations described above, producing a total of 72 sentences. A sample set of the variants is provided in (1); see the *Experimental items* subsection below for the full list of experimental items. The 72 sentences were then distributed into three lists in a Latin square fashion, such that in each list, each frame appeared only once and there were eight items for each condition.

- (1) a. *No subject pronoun, only S-V agreement present*
- (i) Hana mësła, że zakochała-**m** sã w sąsadze.
Hana thought that fell.in.love.L.SG.F-1SG SE in neighbour [K, artificial]
- (ii) Hania myślała, że zakochała-**m** się w sąsiedzie.
Hania thought that fell.in.love.SG.F-1SG SE in neighbour [P]

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- (iii) Hania myślała, iże-**ch** sie zamiyłowwała we sōmsiedzie.
 Hania thought that-1SG SE fell.in.love.SG.F in neighbour [S]
 ‘Hania thought that I had fallen in love with a neighbour.’
- b. *Subject pronoun and S-V agreement present*
- (i) Hana mësła, że **jô** sã zakòchała-**m** w sąsadze.
 Hana thought that I SE fell.in.love.L.SG.F-1SG in neighbour [K, artificial]
- (ii) Hania myślała, że **ja** zakochała-**m** się w sąsiedzie.
 Hania thought that I fell.in.love.SG.F-1SG SE in neighbour [P]
- (iii) Hania myślała, iże **jô** sie zamiyłowwała-**ch** we sōmsiedzie.
 Hania thought that I SE fell.in.love.SG.F-1SG in neighbour [S]
 ‘Hania thought that I had fallen in love with a neighbour.’
- c. *Only subject pronoun present, no S-V [PERS/NUM] agreement*
- (i) Hana mësła, że **jô** sã zakòchała w sąsadze.
 Hana thought that I SE fell.in.love.L.SG.F in neighbour [K]
- (ii) Hania myślała, że **ja** zakochała się w sąsiedzie.
 Hania thought that I fell.in.love.SG.F SE in neighbour [P, artificial]
- (iii) Hania myślała, iże **jô** sie zamiyłowwała w sōmsiedzie.
 Hania thought that I SE fell.in.love.SG.F in neighbour [S]
 ‘Hania thought that I had fallen in love with a neighbour.’

A set of 24 grammatical and 32 ungrammatical filler sentences were created. Ungrammatical fillers contained [SG/PL] subject-verb agreement mismatches (16 fillers) or incorrect case value (16 fillers). We picked this distribution of grammatical and ungrammatical fillers based on our expectations for the P target items. Specifically, we expected 16 target items to be acceptable and 8 target items to be unacceptable. This distribution of fillers would then ensure that each participant saw roughly as many acceptable as unacceptable sentences. This in turn would discourage participants from artificially adjusting their responses so that they can distribute responses uniformly across the rating scale.

The filler sentences were added to each list of target sentences. We used the PCIBex experimental platform (Zehr and Schwarz 2018) to pseudo-randomise the items, so that no two target items appeared consecutively.

Since K and S are mostly spoken, all materials were auditory. Native speakers of K, P, and S were recorded reading our materials aloud; they were instructed to produce the ungrammatical sentences with intonational patterns typically used for grammatical sentences.

Participants

We recruited 40 participants for each language, by word of mouth. Because K and S speakers are in general proficient in P, we did not impose any exclusion criterion based on the languages they knew. Participants received gifts or vouchers comparable in value as compensation.

Procedure

The experiments were conducted with research assistants who are native speakers of K, P, or S. Following gathering informed consent, the experimental sessions started with a short informal conversation in the target language, so that participants would feel more comfortable in a formal experimental situation. In addition, doing so helped to mitigate the risk of the intrusion of P in the K and S sessions.

The research assistants explained the instructions in an informal manner in the target language. In particular, participants were asked to determine whether the sentence they hear could be produced by a native speaker of the language (e.g., “Does the sentence sound weird, as if it were spoken by someone who

did not grow up speaking S, or is it natural?") by selecting an option on the scale from 1 (very bad) to 7 (very good) on a laptop keyboard. Recordings were played from a laptop using PCIBex. Participants could listen to each recording as many times as they preferred before they gave a response.

Data analysis

Due to unforeseen data loss, we only collected responses from 38 K participants and 31 S participants. We applied the z-score transformation to each participant's ratings, in order to control for differences in how the participant used the rating scale.

Filler items were also analyzed as a way to exclude participants who might have not been paying attention. Within each language, we calculated a mean acceptability and standard deviation (in z-score terms) for each filler item. We then checked whether each participant's acceptability rating for that filler item was beyond two standard deviations of the mean acceptability; we consider such a response to be "extreme" and atypical. We only include participants who gave six or fewer "extreme" responses: 34 P participants, 31 K participants, and 29 S participants.

Results and discussion

Mean acceptability ratings for the three conditions are displayed in Figure 1. In P, conditions with overt [PERS/NUM] agreement auxiliary clitics are rated high (z-score above 0.8); the absence of overt [PERS/NUM] agreement is rated much worse (z-score of -0.7). The reverse is true for K: the only condition that received a high rating was the one without overt [PERS/NUM] agreement. We fitted a linear mixed effects model of acceptability ratings with the three conditions and language as predictors, random intercepts for participants and item, and by-item random slopes for condition (models with more complex random effects were attempted but yielded singular fits or did not converge). This model confirmed that the two overt [PERS/NUM] agreement conditions are more acceptable in P (for "no pronoun, agreement": $b = 1.66$, $s.e. = 0.12$, $t = 14.1$, $p < .01$; for "pronoun, agreement": $b = 1.52$, $s.e. = 0.12$, $t = 13.0$, $p < .01$), and also found significant interaction effects between the conditions and K, confirming that K and P are mirror images (interaction effect for "no pronoun, agreement": $b = -2.79$, $s.e. = 0.17$, $t = -16.7$, $p < .01$; for "pronoun, agreement": $b = -2.75$, $s.e. = 0.17$, $t = -16.5$, $p < .01$).

Figure 1: Acceptability of overt/null [PERS/NUM] agreement with overt/null 1st and 2nd person pronoun subjects

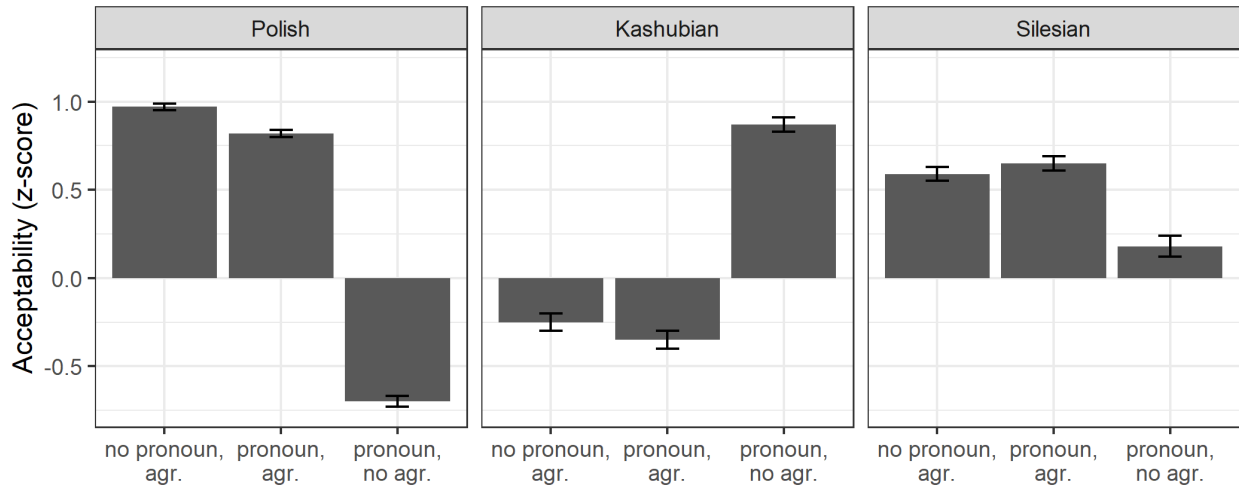
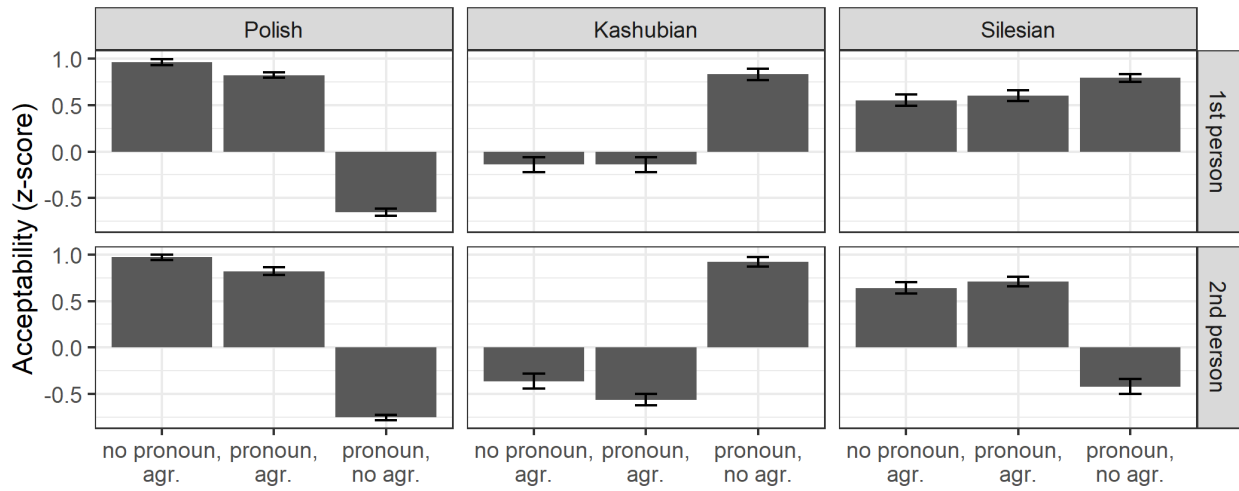


Figure 1 also suggests that S overall resembles P, in the sense that sentences with overt [PERS/NUM] agreement are more acceptable. But as Figure 2 shows, the underlying picture is more complex. Confirming our speculations, a first person agreement auxiliary can be absent – sentences with or without first person agreement auxiliaries are rated as acceptable in S (z-scores above 0.5), while the absence of a second person agreement auxiliary is unacceptable in both P and S (both z-scores negative). We fitted a linear mixed effects model to the S acceptability data, with subject/agreement combinations and person (first vs. second person) as predictors, along with random intercepts for participant and item, and random slopes for subject/agreement combinations. In particular, there was a significant acceptability difference between first and second person for the “pronoun, no agreement” combination ($b = -1.19$, $s.e. = 0.12$, $t = -9.81$, $p < .01$).

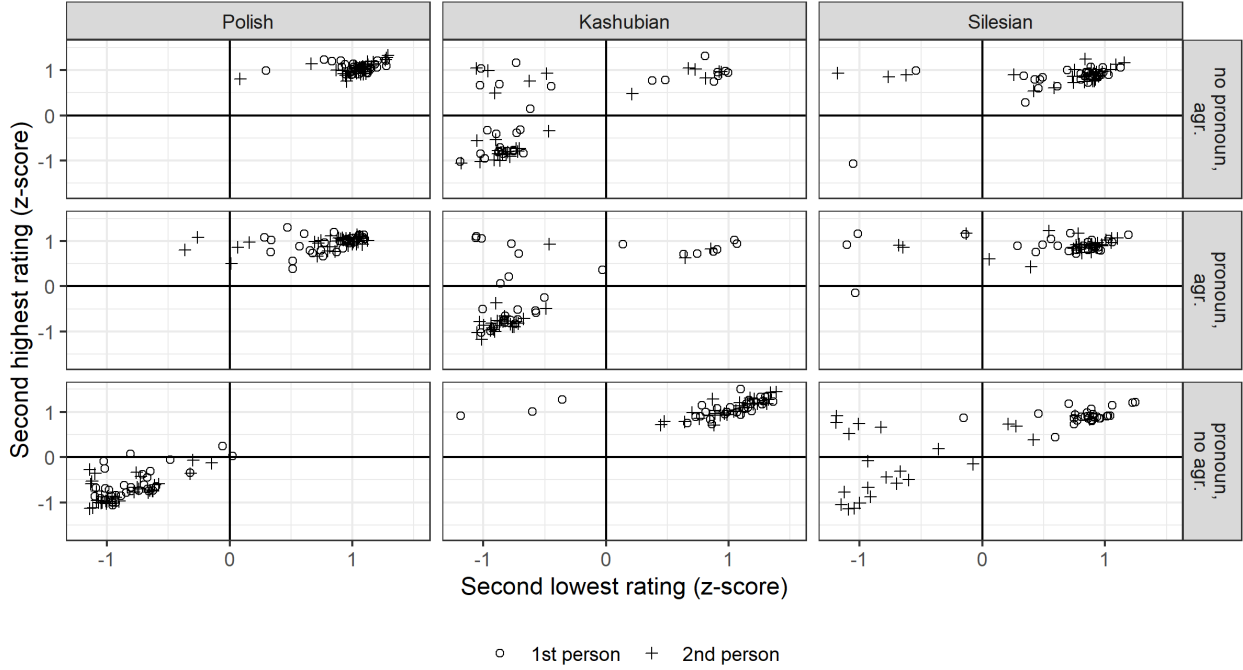
Figure 2: Acceptability of overt/null [PERS/NUM] agreement with overt/null pronoun subjects, by first and second person



In addition, we ran a post-hoc analysis to determine the extent of variation within K and S (cf. [Pañeda and Kush 2021](#); [Huang 2024](#)). For each condition, we extracted and sorted each participant’s ratings for sentences with 1st person subjects. We then dropped the top and bottom ratings, on the assumption that these might be outliers. The remaining second-highest and second-lowest ratings were then plotted for each participant. Plotting the second-highest and second-lowest ratings provide a visual indicator of how consistently that participant rated these sentences: participants appearing in the top right quadrant in the plots in Figure 3 consistently accepted these sentences (z-scores above zero); those in the bottom left quadrant consistently rejected them (z-scores below 0); and those in the top left were inconsistent raters. We repeated the same analysis, this time for sentences with 2nd person subjects.

As Figure 3 shows, there is little internal variation in P (left column): participants consistently accepted sentences with overt [PERS/NUM] agreement auxiliaries and rejected sentences without these auxiliaries. K (middle column) still appears to be the mirror image of P, in that only a minority of participants consistently accepted sentences with overt agreement auxiliaries, while a vast majority of participants consistently accepted sentences with no agreement auxiliaries. As for S (right column), the overall distribution of participant responses resembles that of P, with an important exception in sentences without agreement auxiliaries. For these sentences in S, participants consistently accepted them only when the subject was in the 1st person (circles in the bottom right plot), but not when the subject was in the 2nd person (crosses). Overall, these results show that K and S exhibit relatively little within-language variation for [PERS/NUM] agreement auxiliaries, even though they are minority languages with otherwise substantial dialectal vari-

Figure 3: Variation in judgments across participants



ation.

Finally, we note that our data also suggest the existence of more subtle acceptability contrasts. For example, in P, in the presence of an overt [PERS/NUM] agreement auxiliary, sentences with a null subject appear to be more acceptable than sentences with an overt pronoun subject (Figure 1). Such contrasts are not predicted by our account or, for that matter, standard theories of the null subject parameter. We believe that these contrasts might reflect differences in the information structure properties of overt and null subjects, but defer a fuller explanation to future research.

Nevertheless, this experiment was able to confirm microvariation in the expression of [PERS/NUM] agreement in K, P, and S. The results validated the existing claims that Central K does not allow overt first and second person agreement auxiliaries in the past tense, unlike P. We also established that S allows either an overt first person agreement auxiliary or its omission, unlike P or K. S also requires the second person agreement auxiliary to be overt, like P and unlike K.

Experimental items

The sentences in (a), (b), and (c) are in Polish, Kashubian, and Silesian respectively. The sentences all feature overt pronouns and [PERS/NUM] agreement auxiliaries. To create the null subject or null [PERS/NUM] agreement sentences, simply omit the overt pronouns (*ja/jô/jõ* ‘I’; *my/mě/my* ‘we’; *ty/tě/ty* ‘you.SG’; *wy/wa/wy* ‘you.PL’) or agreement auxiliaries (see Tables 2 and 3 in the main paper).

- (2)
 - a. Gabi powiedziała, że ja wypila-m rano dwie kawy.
 - b. Gabi rzekła, że jô wëpila-m reno dwie kawë.
 - c. Gabi pedziała, iże jô wypiyła-ch rano dwie kawy.
‘Gabi said that I had had two coffees in the morning.’
- (3)
 - a. Grażyna myślała, że ja zaczęła-m uczyć się francuskiego.
 - b. Grażina mësła, że jô sã zaczęła-m ùczëc francëszczégò.

- c. Grażyna myślała, iże jō zaczął-ch sie uczyć francuskigo.
'Grażyna thought that I had started to learn French.'
- (4) a. Halina powiedziała, że ja znalazła-m portfel w pociągu.
b. Léna rzekła, że jō znalazła-m miészki w banie.
c. Halina pedziała, iże jō znōdła-ch portmanyj we cugu.
'Halina said that I had found my wallet on the train.'
- (5) a. Hania myślała, że ja zakochała-m się w sąsiedzie.
b. Hana mësła, że jō sã zakòchała-m w sąsadze.
c. Hania myślała, iże jō sie zamiyłowala-ch we sòmsiedzie.
'Hania thought that I had fallen in love with my neighbor.'
- (6) a. Hela powiedziała, że ja kupiła-m kwiaty na targu.
b. Léna rzekła, że jō kùpiła-m kwiatë na tòrgù.
c. Hela pedziała, iże jō kupiyla-ch kwiòtki na tòrgu.
'Hela said that I had bought flowers at the market.'
- (7) a. Irena myślała, że ja wysłała-m list do wujka.
b. Jirëna mësła, że jō wësłała-m lëst do wùje.
c. Iryna myślała, iże jō wysłała-ch listek do ujka.
'Irena thought that I had sent a letter to my uncle.'
- (8) a. Karolina myślała, że my tańczyli-śmy całą noc.
b. Karolëna mësła, że më tańcowalë-smë całą noc.
c. Karolina myślała, iże my-chmy tańcowali cołkò noc.
'Karolina thought that we had been dancing all night.'
- (9) a. Iwona powiedziała, że my pomodlili-śmy się przed kolacją.
b. Jiwóna rzekła, że më pòmòdlëlë-smë sã przed wieczera.
c. Iwóna pedziała, iże my-chmy porzykali przed wieczierzòm.
'Iwona said that we had prayed before dinner.'
- (10) a. Jadwiga myślała, że my umyli-śmy samochód na myjni.
b. Jadwiga mësła, że më ùmëlë-smë àuto w mijnie.
c. Jadwiga myślała, iże my-chmy umyli auto na myjni.
'Jadwiga thought that we had washed the car at the car wash.'
- (11) a. Janina powiedziała, że my pracowali-śmy do czternastej.
b. Janina rzekła, że më robili-smë do drëdżi pò pòłnim.
c. Janina pedziała, iże my-chmy robiyli do sztyrnostyj.
'Janina said that we had worked until 2 p.m.'
- (12) a. Jola myślała, że my pili-śmy wodę ze strumienia.
b. Jola mësła, że më pili-smë wòdã ze strëdżi.
c. Jola myślała, iże my-chmy piyli wodã ze strumiynia.
'Jola thought that we had drunk water from the stream.'
- (13) a. Julia powiedziała, że my zbierali-śmy grzyby w lesie.
b. Juliô rzekła, że më zbiërelë-smë grzëbë w lese.
c. Julia pedziała, iże my-chmy zbiyrali grzyby w lesie.
'Julia said that we had picked mushrooms in the forest.'
- (14) a. Kasia myślała, że ty przeprowadził-es się do Anglii.
b. Kasza mësła, że të przecygnał-es do Anglie.
c. Kasia myślała, iże ty-ś sie przekludziòł do Anglije.
'Kasia thought that you had moved to England.'

- (15) a. Klara powiedziała, że ty posadził-eś róże w ogrodzie.
 b. Klara rzekła, że tē pòsadzył-es róże w ògrodze.
 c. Klara pedziała, iże ty-ś posadziół róże we zygodzie.
 'Klara said that you had planted roses in the garden.'
- (16) a. Krysia myślała, że ty ugotował-eś zupę pomidorową.
 b. Krisza mësła, że tē ùgòtowół-es pòmidorową zupã.
 c. Krysia myślała, iże ty-ś uwarzół tōmatowō zupã.
 'Krysia thought that you had cooked tomato soup.'
- (17) a. Lucyna powiedziała, że ty poszedł-eś do dentysty.
 b. Lucyna rzekła, że tē pòszedł-es do dentistë.
 c. Lucyna pedziała, iże ty-ś poszoł do dyntysty.
 'Lucyna said that you had gone to the dentist.'
- (18) a. Magda myślała, że ty okłamał-eś wczoraj policjanta.
 b. Magda mësła, że tē òcëganił-es wczora szandara.
 c. Magda myślała, iże ty-ś ôszydziół wczorej policista.
 'Magda thought that you had lied to the policeman yesterday.'
- (19) a. Gosia powiedziała, że ty ukradł-eś tysiąc dolarów.
 b. Gréta rzekła, że tē ùkròdł-es tèsac dolarów.
 c. Gosia pedziała, iże ty-ś porwoł tysiōnc dolarów.
 'Gosia said that you had stolen a thousand dollars.'
- (20) a. Mariola myślała, że wy składali-ście meble cały wieczór.
 b. Mariola mësła, że wa skłòdal-ësta méble całi wieczór.
 c. Mariola myślała, iże wy-ście skłòdali meble còłki wieczór.
 'Mariola thought that you had been assembling furniture all evening.'
- (21) a. Marlena powiedziała, że wy zapomnieli-ście naładować telefon.
 b. Marlena rzekła, że wa zabòczël-ësta naładowac telefón.
 c. Marlyna pedziała, iże wy-ście zapòmnieli naładować telefôn.
 'Marlena said that you had forgotten to charge your phone.'
- (22) a. Marta myślała, że wy okryli-ście się kocem.
 b. Marta mësła, że wa sã òkrëłë-sta deką.
 c. Marta myślała, iże wy-ście sie ôkryli dekom.
 'Marta thought that you had covered yourselves with a blanket.'
- (23) a. Marzena powiedziała, że wy oglądali-ście film na laptopie.
 b. Marzena rzekła, że wa òbzéralë-sta film na laptopie.
 c. Marzyna pedziała, iże wy-ście patrzeli na film na laptopie.
 'Marzena said that you had been watching a movie on your laptop.'
- (24) a. Mirka myślała, że wy wygrali-ście turniej szachowy.
 b. Mirka mësła, że wa wëgralë-sta szachòwą rozgriwkã.
 c. Mirka myślała, iże wy-ście wygrali szachowy turniuj.
 'Mirka thought that you had won a chess tournament.'
- (25) a. Monika powiedziała, że wy poszli-ście na spacer do parku.
 b. Mònika rzekła, że wa pòszli-sta na szpacér do lasu.
 c. Mōnika pedziała, iże wy-ście poszli na szpacyr do parku.
 'Monika said that you had gone for a walk in the park.'

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